



Harnessing the Potential of Artificial Intelligence: Affordances, Constraints, and Strategic Implications for Professional Services

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ABSTRACT

While Artificial Intelligence (AI) promises to transform business models and enable novel service propositions, it also ushers in intricate managerial challenges. To comprehend AI's disruption as well as harness its potential, we conducted a developmental review of 92 scholarly works centred on financial auditing, the chosen professional service domain. Drawing on the corpus of literature, we applied the Technology Affordance and Constraints Theory (TACT) to dissect the opportunities and challenges that AI introduces to professional services. Our analysis led to conceptualising AI's affordances and constraints for individual practitioners, firms, and professional associations within the auditing domain. Subsequently, we compared these with the affordances and constraints of typical non-AI, classic IT tools in the same domain, highlighting the disruptive essence of AI. Our findings further revealed the reciprocal and tensional relationship between AI affordances and constraints, where actualising AI affordance at the firm level may impose constraints on individual practitioners. Based on our findings, we argue for integrating AI into decision-making by practitioners, propose a dynamic view of task automation and augmentation for firms, and urge an AI-calibrated knowledge system for professional associations. This study contributes to both theory and practice, laying the foundation for a more nuanced discourse on the strategic significance of harnessing AI in professional services.

Introduction

The integration of Artificial Intelligence (AI) technology has significantly disrupted many occupations, including both highly skilled roles and manual labour positions. The professional services sector, characterised by high knowledge intensity and highly idiosyncratic processes, was once considered invulnerable to the disruption of AI and automation (Maister, 1993). However, with the recent advancements in AI technology, this prevailing belief has been challenged and poised for revision (Sampson, 2021; Spring et al., 2022; Susskind & Susskind, 2015). AI has demonstrated superior capabilities in areas akin to human cognition, such as learning, interaction, and problem-solving (Coombs et al., 2020). Further bolstered by the burgeoning availability of vast data, refined algorithms, and formidable computational power, AI has rapidly evolved into a fundamental component in the delivery of professional services.

The recent advancements in AI have ignited a surge of discussion. We acknowledge that the current body of Information Systems (IS) literature is well-equipped to establish the definition of AI (Berente et al., 2021; Collins et al., 2021), explore the strategic implications of AI on service, knowledge, and organisational learning (Alavi et al., 2024; Coombs et al., 2020; Sturm et al., 2021), as well

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as investigate the interplay between AI and business strategy (Borges et al., 2021; J. Li et al., 2021). Nevertheless, the present state of AI research is characterised by a siloed approach, wherein the technical aspect solely concentrates on AI technology features while the social aspect only emphasises the impact of AI on organisations and individuals (Berente et al., 2021). As AI continues to evolve in various industries, its implications extend far beyond simple technical enhancements. It fosters innovative ways of conducting business, catalysing the development of novel service propositions, and propelling organisational transformation (Kane et al., 2019; Raisch & Krakowski, 2021). However, with these promising opportunities comes a suite of intricate challenges that necessitate thoughtful management, such as black box problems and accountability issues (Barredo Arrieta et al., 2020; Miller, 2019). In the context of professional services, the strategic importance of harnessing AI's opportunities while mitigating its associated challenges for individuals, organisations, and the entire profession cannot be overstated.

The societal impact of AI is a dynamic interplay between its technological capabilities and human societal frameworks, contingent upon balancing automation with augmentation, aligning physical and digital realities, and preserving human decision-making processes (Sabherwal & Grover, 2024). To further investigate this impact, we adopted a socio-technical perspective, looking to align the social dimension of AI (i.e., the human intentions and consequences of AI interaction) with its technical essence (referring to AI's unique capabilities, such as its learning, automation, and cognitive-like functions). To effectively merge socio-technical thinking into our research, we utilised the Technology Affordance and Constraints Theory (TACT). TACT provides a comprehensive lens through which we can perceive the interplay between the goal orientation of social actors and the properties of technology, leading to the emergence of affordances and constraints (Faraj & Azad, 2012; Majchrzak & Markus, 2012). Consequently, TACT becomes an appropriate tool to address our central concern: delineating the potential benefits and challenges that AI presents to professional services.

Our employment of TACT, focusing on both the properties of AI and actors' goal orientation, helps explain how AI becomes integrated into and impacts action for individual practitioners, firms, and professional associations. This lens views AI not as a stand-alone entity, but as an integral element that enables certain types of action while constraining others. TACT further enables and directs us to ask more specific and precise questions (Faraj & Azad, 2012; Leonardi & Vaast, 2017). For example, instead of asking how AI affects individual decision making, we can probe how the affordances or constraints of AI become intertwined in producing the output of decisions.

Furthermore, the analysis of affordances and constraints serves as the foundation to understand AI's implications for entities within professional service settings. The implications of AI for a particular entity, such as practitioners, should not be analysed in isolation. It is essential to consider how other entities, such as firms and professional associations within the broader environment, interact with AI as well. AI may support one entity but constrain others. For example, while AI affords firms the opportunity to automate processes, it may constrain practitioners by jeopardising job security. Therefore, analysing the alignment and conflict of affordances and constraints across different entities can help us articulate the likely outcomes and implications for practitioners, firms, and professional associations when they encounter AI. We accordingly developed the following research questions:

RQ1: How do the affordances and constraints of AI impact the actions and outcomes for individual practitioners, firms, and professional associations within professional service settings, and how do these effects interact across these entities?

In our quest to better understand the intricacies of AI, we argue that it is essential to compare the affordances and constraints of AI with those of traditional IT tools. This comparative approach not only sheds light on AI's disruptive nature, but also forms the basis to further discuss its strategic implications. When we compare AI's affordances (such as its ability to automate processes and augment decision-making) with the more static and deterministic solutions of classic IT tools, the need for a revised approach to technology use and management becomes evident. The divergence in affordances and constraints between AI and classic IT demands a paradigm shift, pushing us to rethink our strategies, methodologies, and mindsets. In light of this argument, we pose our second research question:

RQ2: How do the affordances and constraints of AI contrast with those of the classic IT tools?

This study adopts a literature review method to address the research questions. The review follows the developmental review principles proposed by Templier and Pare (2015). Using TACT as the theoretical lens to synthesise knowledge from literature has become a widespread practice within the IS discipline. This approach has been applied in various studies, such as examining impression management strategies in enterprise social media platforms (Sun et al., 2021) and analysing the affordances of social media for organising (Leonardi & Vaast, 2017).

TACT theorists cautioned that the study of affordances and constraints must be situated within specific contexts and boundary conditions (Burton-Jones & Volkoff, 2017; Strong et al., 2014). In extant literature, TACT is usually used as a mid-range theory to study the interaction between technology and social actors in a particular context (Leidner et al., 2018; Tim et al., 2020). While acknowledging the rich discussions around AI applications across a spectrum of professional service industries, including accounting (Moll & Yigitbasioglu, 2019), legal (Spring et al., 2022), and healthcare practices (Yu et al., 2018), we understood that it is an arduous task to effectively consolidate findings in a single review and theorise AI's affordances and constraints for such a diverse range. Therefore, striking the right balance in a research context is paramount; while it is essential to avoid overly broad or vague scopes, it is equally important not to narrow the focus excessively. In our study, we tactically focused on the auditing industry, exemplifying its role within professional services to illuminate the considerable impact of AI. Auditing is a professional service that assures the reliability of financial reports prepared by management, ensuring information accurately reflects the company's financial position and performance (Knechel et al., 2013). This professional engagement requires auditors to conduct a systematic examination of financial records, transactions, and controls within a company to assess their accuracy, completeness, and compliance with accounting standards and regulations.

Our selection of auditing has two main considerations. First, AI's influence on auditing profession is prominently observed. A wide range of AI applications has been integrated into auditing practices, from traditional expert systems to advanced machine learning

analytics, natural language processing, and machine vision (Munoko et al., 2020; Sun, 2019; Sutton et al., 2016). These AI applications have profoundly disrupted and revolutionised the auditing profession and practices (Goto, 2022, 2023; Yang et al., 2024). Second, auditing practice is highly convergent around the world, which reduces concerns about practice divergences across the world that could bias research outcomes. Despite the wide array of economic, technological, cultural, and regulatory differences globally, the harmonisation of auditing practices has significantly advanced (Boolaky & Soobaroyen, 2017; Gangolly et al., 2002). Driving auditing harmonisation is the global accounting governance body, the International Federation of Accountants (IFAC), along with its member associations such as the AICPA in the US and JICPA in Japan (Anandarajan & Kleinman, 2021). To date, over 100 countries have adopted the International Standards on Auditing (IFAC, 2019). Regulatory bodies from leading economies, including the Public Company Accounting Oversight Board in the United States and the Public Oversight Board in the United Kingdom, collaborate to enhance auditing quality through convergence efforts. Economically significant nations like Brazil, India, and China are also endeavouring to synchronise their auditing standards with global norms (Kleinman et al., 2014). These efforts are supported by educational and training programs designed to align with international standards and by regulatory and legislative changes aimed at bolstering corporate governance (Francis, 2011). Collectively, these developments foster a globally harmonised auditing practice, making auditing an ideal research context. While the research context is AI application in auditing, it is important to note that auditing and other professional services share fundamental characteristics, such as high knowledge intensity, low capital intensity, and a professionalised workforce (Spring et al., 2022; Susskind & Susskind, 2015; von Nordenflycht, 2010). Therefore, our in-depth exploration of the auditing industry is not isolated. Rather, it establishes a foundation, allowing us to extrapolate our findings and interpretations to a broader professional service context.

The paper is structured as follows. In the next section, we provide the background of AI applications in the auditing industries and an overview of the TACT. Next, we present the study's methodological approach to conducting a developmental review. This is followed by findings that synthesise and critically analyse extant literature. Finally, we elaborate on the study's implications and contributions.

Research background

The application of AI in auditing

Given that the auditing industry is the focal research context, it becomes vital to provide a concise yet comprehensive overview of the use of AI within this sector. To ensure a nuanced understanding, we must first establish our operational definition of AI in the context of this study. We anchored our understanding on the most cited definition provided by Russell et al. (2010), that AI “enables the machine to exhibit human intelligence, including the ability to perceive, reason, learn, and interact, etc (p. 31, 44, 50)”. We also considered the common typologies of AI, including machine learning, natural language processing, robotics, machine vision, and expert systems (Collins et al., 2021).

Researchers and practitioners began discussing the potential impact of AI on audit methodologies as early as the 1980 s (Elliott, 1986). In these initial stages, AI was commonly associated with expert systems and decision support systems (Elam & Mead, 1990; Elliott, 1986). These technologies were lauded for their intelligence, given their capacity to employ decision rules to delineate complex systems and interact coherently with auditors. However, their application, alongside automation, was primarily restricted to structured and semi-structured tasks (Abdolmohammadi, 1991).

In those early days, the professional understanding of AI was somewhat limited but continued to evolve alongside technological advancements. During the late 1990 s and early 2000 s, opportunities arose to utilise artificial neural network software such as Neuro Shell 2, aiding in the classification and prediction of management fraud and bankruptcy (Charitou et al., 2004; Green & Choi, 1997; Hian & Chan, 2004). Research in this area flourished in the 2010 s, with various machine learning algorithms being tested with diverse parameters and training samples to enhance classification accuracy. These algorithms encompassed artificial neural network, logistic regression, Bayesian networks, decision tree models, support vector machines, and ensemble methods (Hajek & Henriques, 2017; Perols, 2011).

With technology maturing, the evolution in the scope of data processed by machine learning is striking. Initially dealing with just five financial ratios (Green & Choi, 1997), it now capably handles an expansive dataset comprising 84 financial ratios and 21 categories of social media and Management Discussion and Analysis (MD&A) content (Dong et al., 2018). This content encompasses sentiment, topical, emotional, linguistic, and social network features. The versatile data handling capacity of AI has led to its expanded application in auditing. Beyond the scope of expert systems, decision support systems, and algorithmic software, the AI application horizon now includes advanced technologies such as drones, machine vision, and natural language processing (Tiron-Tudor & Deliu, 2022). Simultaneously, the industry has transitioned from using stand-alone AI applications to incorporating them into comprehensive smart auditing platforms, such as Deloitte's ‘Argus’ and PwC's ‘GL.ai’. This shift underscores the increasing sophistication of the AI ecosystem in auditing.

An example of non-AI classic IT tools used in auditing

Articulating what constitutes non-AI classic IT can be as challenging as defining AI, primarily due to the differences in the level of ‘intelligence’ they possess. While AI is known for its ability to mimic and simulate human intelligence and decision-making processes, non-AI classic IT is more rudimentary in its function, executing only the explicit instructions programmed into it. In essence, non-AI classic IT does not exhibit cognitive functions traditionally associated with the human mind, such as perception, reasoning, learning,

interaction with the environment, problem-solving, decision-making, and creativity (Rai et al., 2019).

Turning our focus to the auditing industry, a representative category of non-AI classic IT is manifested in the form of Computer-Assisted Auditing Tools and Techniques (CAATTs). This umbrella term encapsulates a range of general-purpose software such as spreadsheet and word processors, and generalised auditing software such as CaseWare IDEA (Ahmi & Kent, 2013). The term CAATT was first created in the 1980 s, when the auditing profession had just popularised computers. Professionals and researchers use this term to distinguish between paper-based auditing and computer-assisted auditing. Initially, researchers prototyped Audit Command Language to consolidate the audit support functions into a common language of management information systems (Will, 1983). Subsequently, CAATTs have been extensively utilised in facilitating auditing tasks, notably in detecting material misstatements, control deficiencies, and fraud (Bradford et al., 2020). Indeed, prior to the advent of contemporary AI solutions, they were arguably the most vital IT tools used by auditors.

We regard CAATTs as non-AI, classic IT – considering their operational capabilities are tethered to their pre-programmed functions, such as data analysis and report generation. Consequently, these tools cannot transcend their predefined tasks, nor are they equipped to learn from data or make data-driven decisions. These features distinguish them from AI, which has a higher level of intelligence and autonomy.

Technology affordance and constraints theory (TACT)

We adopted TACT to answer our research questions considering its power in explaining the process and consequence of the human-IT interaction at multiple societal levels. The concept of affordance, initially introduced in ecological psychology by Gibson (1977), describes the relationship between species and the environment as “the affordances of the environment are what it offers the animal, what it provides or furnishes, either for good or ill” (p. 127). Subsequently, the concept gained attention in other fields including IS. Hutchby (2001) was one of the first to apply the affordance and constraint concept in understanding the complex relationship between users and IT artefacts.

We draw on prior literature and adopt the following definition of affordances and constraints: “The concept of technology affordance refers to an action potential, that is, to what an individual or organisation with a particular purpose can do with a technology or information system; technology constraint refers to ways in which an individual or organisation can be held back from accomplishing a particular goal when using a technology or system” (Majchrzak & Markus, 2012, p. 1). That means affordances and constraints exist as a relationship between goal-directed social actors and artefacts, where artefacts enable and constrain potential actions to achieve goals. Pozzi et al. (2014) explained the relationship by making a good example of an actor who desires to enter a room by passing through a door. The door’s affordance is determined by its width and the actor’s physical capability to pass through it. However, the door may become a constraint for an actor whose width exceeds the door size. IS scholars have widely accepted TACT. For example, in the context of tensions in crowdsourcing for innovation, TACT has been used to identify design affordances and constraints that can increase the possibility of co-creation among strangers (Majchrzak & Malhotra, 2013). In other studies, TACT has helped researchers address assumptions about the differentiation between innovation process and outcome, specifically addressing complex, dynamic interactions and the duality of digital innovation (Nambisan et al., 2017).

We posit that TACT provides a robust theoretical framework for addressing our research questions, which seek to articulate AI affordances and constraints and compare them with classic IT. We are guided by prior research advocating the use of affordances as a theoretical lens for studying IS phenomena (Faraj & Azad, 2012). A typical Cartesian perspective may differentiate between classic IT and AI by characterising the features and functions of the two types of artefacts. However, IS scholars have emphasised the risk of viewing technology as a bundle of features and product classes, which leaves researchers unable to distinguish between taken-for-granted features and emergent technology-in-use when users encounter IT in a situated context (Faraj & Azad, 2012). For example, a designed feature of ChatGPT might be natural language generation based on a given prompt. However, it has the emergent property to produce novel ideas and synthesise knowledge during its interaction with users. This emergent property goes beyond what might be considered as taken-for-granted features possessed by other natural language processing tools. Researchers who view ChatGPT as a bundle of features may overlook the emergent affordances that distinguish it from other artefacts. Therefore, we adopt TACT to overcome such limitations. The analysis of affordances and constraints is rooted in the mutual relations between social actors and artefacts.

Studying AI through the lens of TACT enables us to understand its inherent role in shaping actions taken by social actors. AI, with its advanced capabilities, offers the possibility of affording certain types of actions that would have been challenging or impossible to achieve with non-AI, classic IT (Faraj & Azad, 2012; Leonardi & Vaast, 2017). On the other hand, it can also introduce constraints, limiting other desirable actions (Leonardi, 2011; Majchrzak & Markus, 2012). However, as suggested by Leonardi and Vaast (2017), this perspective does not assign AI an overly dominant role. Rather, it positions AI as a facilitator, highlighting that the actions of the social actors are primary, and the role of AI, with its affordances and constraints, supports these actions as enabling conditions.

Applying this TACT lens, we delve into the literature to articulate the goal orientation of social actors at three levels within auditing industry: the individual practitioners, firms, and the professional associations. We then highlight the properties of various types of AI applications used in auditing, contrasting these with the properties of CAATTs. This approach permits us to conceptualise the unique affordances and constraints of AI relative to classic IT. This comparative analysis thus sets the stage for our ensuing discussion on the strategic actions that individuals, firms, and the professional association should consider when harnessing the potential of AI and mitigating its constraints. It provides a foundation for recommending adaptive measures and strategies that allow actors to effectively navigate the evolving landscape of professional services shaped by AI.

Table 1
Literature Selection Process.

Steps	Tasks	Sub-tasks
Define	Define criteria for inclusion and exclusion	Quality threshold: Journals ranked 4*, 4, or 3 on the ABS list 2021 or A* or A journals on the ABDC list 2019. Inclusion criteria: articles must discuss either CAATTs or AI used in financial statement auditing. Exclusion criteria: articles 1) discuss other types of technology; and 2) context other than external financial statement auditing
	Define field of research	Accounting, Management, and Information Systems
	Determine sources	Scopus Meta Database
	Decide specific search terms	AI artefacts: (“AI” OR “artificial intelligence” OR “machine learning” OR “neural networks” OR cognitive* OR automation* OR robot* OR augment* OR “Deep Learning” OR “Represent* Learning”) AND (audit OR auditing OR auditor)). CAATTs: (“Generalised Audit Software” OR “Computer Assisted Audit Tools and Techniques” OR “CAATTs” OR “Computer Assisted Audit Techniques” OR “CAATTs” OR “Computer Audit Software” OR “Computer Audit Package” OR “audit command language” OR “ACL” OR “interactive data extraction and analysis” OR “Test Data” OR “integrated test facility” OR “Utility program” OR “Specialised audit software”) AND (audit OR auditing OR auditor))
Search	Conduct search	Search period: unrestricted Search execution: April 2022 Results: 302 CAATTs papers 5608 AI papers
Select	Refine sample	Applying the ABS and ABDC quality filters: 39 CAATTs papers and 319 AI papers. Reading the Title, Abstract and Full Text: 23 CAATTs papers and 58 AI papers. Forwards and Backwards search: 3 new CAATTs papers and 8 new AI papers The final pool: 26 CAATTs papers and 66 AI papers

Methodology

To address our research questions, we conducted a developmental review (Templier & Pare, 2015). This approach is not only rigorous in its structured search strategy but also excels in its concept-centric analysis, blending both conceptual and empirical studies with different methods and designs. One of the pivotal reasons behind opting for this type of review was its proficiency in conceptualising the affordances and constraints of both AI and classic IT. While prior studies did not explicitly articulate these affordances and constraints, they laid foundational insights that allowed us to derive and define the focal concepts with greater clarity. Thus, this method enables us to construct meaningful knowledge from existing studies. The developmental review method has been employed in recent studies to propose novel conceptualisations and theoretical foundations in business model portfolio (Westerveld et al., 2023) and circular economy (Zeiss et al., 2021).

To uphold the standards of rigour, transparency and systematicity, we adhered to the guiding principles of conducting developmental review proposed by Templier and Paré (2015), and made further reference to the pragmatic methodological guidance in related works (Webster & Watson, 2002; Wolfswinkel et al., 2013). Accordingly, our literature selection process was threefold. Initially, we delineated our research fields, pinpointed literature sources, set specific search terms, and established clear inclusion and exclusion criteria. Subsequently, we executed the search protocol to collect articles. In the final phase, we meticulously sifted through these articles based on our earlier defined criteria. A breakdown of our literature selection methods – inclusive of keywords, databases, refinements, and the resultant count – is illustrated in Table 1.

This review applies the quality threshold by referring to two widely recognised journal ranking guides: the Academic Journal Guide 2021 (ABS list) and the Australian Business Deans Council 2019 (ABDC list). We included journals ranked 4*, 4, or 3 on the ABS list or A* or A journals on the ABDC list. Applying such quality thresholds, we only included articles from 357 high-ranking journals discussing either CAATTs or AI used in financial statement auditing. To ensure the currency and comprehensiveness of our literature search, we adopted the search terms from the most recent literature reviews on AI (Collins et al., 2021; Borges et al., 2021), and generated keywords for CAATTs based on seminal articles within the field (Bierstaker et al., 2014; Braun & Davis, 2003; Curtis & Payne, 2014; Janvrin et al., 2008). Our literature search was conducted in April 2022 with the Scopus Database, searching paper titles, abstracts, and keywords with our predefined terms. Without imposing a timeframe restriction, our goal was to capture a complete picture of the evolution and application of CAATTs and AI in auditing. From an initial finding of 5901 articles, and after applying quality filters, reviewing titles and abstracts, as well as conducting backward and forward citation tracing, we distilled our findings to a core collection of 92 papers. Our Appendix provides a summary of the 92 papers, covering their methodologies, main findings, and focal technologies, along with the distribution of publication years and research types.

Our literature analysis began with a thorough review of five randomly selected articles on AI and five on CAATTs. This initial step allowed us to become familiar with the subject matter and build a foundational understanding of the topics. Once grounded in the basics, we moved forward with a more detailed examination, using Nvivo to code a total of 92 papers. Central to our literature analysis was the focus dictated by our research question: the affordances and constraints of AI and CAATTs for individual practitioners, firms, and professional associations. It was thus critical to analyse the fundamental elements constituting affordances and constraints. Drawing inspiration from the affordance literature, we recognised that these notions are anchored in two primary facets: the inherent properties of technology and the goal orientation of actors (Faraj & Azad, 2012; Majchrzak & Markus, 2012). Therefore, the first round of coding began with a deep dive into these foundational elements. During this phase, we pinpointed and inductively coded excerpts that discussed the goal orientations of actors and properties of AI and CAATTs. For instance, we identified the following sentences discussing the function of a natural language processor, and we coded ‘Contextual vectorisation of text’ accordingly: “To make a text

document understandable and machine-readable, a 'shallow' neural network such as Word2vec transforms the document into sets of vectors, termed 'word embeddings', that numerically represent the context of each word within the document (Sun, 2019, p. 94)".

Subsequently, we consolidated the initial codes into parent categories. This was achieved by iteratively revisiting the concepts extracted from the literature and evaluating the interrelations among the first-order codes (Webster & Watson, 2002; Wolfswinkel et al., 2013). This iterative process ensured that these codes were aptly grouped into overarching parent categories. Following the above example, the first-order codes of vectorisation, tokenisation, tagging, parsing, and others are merged into the parent category: functional properties of natural language processor. By the conclusion of the second round of coding, we summarised the properties for five AI technologies and two CAATTs used in auditing contexts. As for another fundamental element, we identified seven categories for goal orientations: two for individual practitioners, three for firms, and two for professional associations.

Finally, we transitioned to the coding of our ultimate themes: affordances and constraints, which involved identifying and cultivating relationships between main categories (Wolfswinkel et al., 2013). The coding of affordances and constraints demanded an iterative comparison of actors' goal orientation with the properties of artefacts, drawing insights from the outcomes associated with tool utilisation (Leidner et al., 2018). The following example indicates how we coded 'automate process', a firm-level action afforded by a natural language processor.

Natural language processing software [artefact] is functional, with semantic and syntax analysis, entity recognition, sentiment analysis, and pattern recognition [functional properties]. In the context of auditing firms, which aim to enhance their service quality and efficiency [firms' goal], these properties can offer several action potentials. Particularly, the software's property of the ability to recognise patterns and analyse semantics can be utilised to automate the process of reviewing contracts, meeting minutes, and management disclosure – ensuring anomalies and inconsistencies are promptly detected [affordance]. Such automation can significantly elevate auditing services by boosting efficiency and effectiveness, thereby enabling firms to achieve their short-term business objectives [outcome].

We meticulously applied the iterative coding process outlined above to comparatively analyse the properties of five types of AI artefacts and two types of CAATTs, aligning with the goal orientations of individual practitioners, firms, and professional associations. As a result, we identified nine affordances and eight constraints, which became the central themes of our findings and subsequent discussion.

Findings

In this section, we first report the findings on the properties of AI and CAATTs, and the goal orientations of individual practitioners, firms, and professional associations. These findings form the basis for further articulating how the affordances and constraints emerge due to the interactions between the properties of artefacts and goal-oriented actors.

The properties of AI and CAATTs

We identified five groups of AI artefacts used in auditing, including robotic process automation (RPA), expert systems (ES), machine learning number analytics (MLNA),¹ natural language processors (NLP), and computer vision (CV). This classification draws from categorisations found in prior literature (Collins et al., 2021; Sutton et al., 2016), and we applied this categorisation to analyse discussions on AI applications in the auditing field. Robotic process automation is an application that allows users to configure one or more scripts (sometimes referred to as "bots") to activate specific keystrokes in an automated fashion. Expert systems contain a knowledge base and inference engine that are designed to mimic the decision-making ability of a human expert. Machine learning number analytics contain a single or a collection of machine learning algorithms that can be used for numerical data analysis and predictive modelling, usually with a graphical user interface that allows researchers to set up and tune the functions and parameters. A natural language processor is a toolkit based on machine learning algorithms that can process and analyse natural language with the capability of 'understanding' the content and contextual nuances. Computer vision refers to a machine that is capable of interpreting and responding to visual inputs, often using cameras, digital signals, and algorithms.

Regarding this categorisation, we acknowledged the difference between strong AI and weak AI discussed in prior research (Collins et al., 2021; Wolfe, 1991). Weak AI artefacts cannot improve their methods and results without human intervention. Instead, they act as assistants to humans by providing recommendations or automating simple processes. They are meant to assist with repetitive decisions and complex problems but not take over cognitive tasks from humans (Coombs et al., 2020). Robotic process automation and expert systems are considered weak AI, mainly because they cannot improve their performance without human intervention. The core property of these weak AI systems is their 'rule-based' nature, which means they operate strictly within the boundaries of explicitly defined rules. By contrast, machine learning number analytics, natural language processors, and computer vision are considered strong AI because machine learning algorithms power them. These artefacts are distinct from weak AI because its 'rule-discovering' property allows it both to follow explicit rules and discover new rules. This categorisation aligns with prior literature discussing the different levels of digital labour, asserting that weak AI primarily contributes to automation while strong AI has more cognitive engagement (Collins et al., 2021; Tiron-Tudor & Deliu, 2022). Table 2 summarises and categorises the AI artefacts used in auditing. Noteworthy,

¹ We use the term 'number' rather than 'data' on purpose. While number, text, image, and even voice can be regarded as data, this category of artefact is only capable of processing numerical data. We differentiate this group of artefacts from others such as natural language processors and computer vision.

Table 2
Categories of AI Artefacts used in Auditing.

Categories	Example artefact	Properties	Exemplar research
RPA	Automation Anywhere; Blue Prism; UiPath	RPA deploys software bots programmed to emulate human interactions with digital systems. Typically anchored in a control dashboard, these bots operate by mimicking front-end user interface actions. They are highly configurable and can interact seamlessly across various information systems, following the mapped business processes. Other properties of RPA include error logging, visualising, scaling, and being non-invasive.	Eulerich et al. (2022) ; Huang and Vasarhelyi (2019) ; Zhang et al. (2021)
ES	Analytical diagnostic tool; Client risk assessment tool; Audit Checklist; Audit expert system	ES contains an explanation facility (explain to users why certain information is needed and how the solution is reached), an inference engine (rule interpreter with an analytical framework for generating advice to users), a knowledge base (a set of IF-THEN rules) and user interface allowing users to give facts about the issue. Their sophisticated reasoning processes can employ strategies like backward and forward chaining, while some systems can passively learn from users' feedback, ensuring their responses remain relevant and accurate.	Abdolmohammadi (1991) ; Baldwin-Morgan (1995) ; Elam and Mead (1990) ; Nelson et al. (2000) ; Sutton et al. (2016) ; Dowling and Leech (2007)
NLP	Word2vec; Stanford CoreNLP toolkit; Wordnet Stemmer; GPT-2;	NLP systems leverage algorithms to understand and process human language. They handle text pre-processing, linguistic annotation, feature engineering, and predictive modelling. Advanced capabilities include semantic understanding, sentiment analysis, entity recognition, machine translation, and powering conversational agents. These tools are pivotal in facilitating natural, human-machine interactions.	Brown et al. (2020) ; Craja et al. (2020) ; Dong et al. (2018)
MLNA	Brainmaker; Neuralyst; Neural Works Professional IV Plus; NeuroShell 2; N-NET; WinNN; InTrees;	MLNA tools are software packages equipped with data mining and machine learning capabilities. Essential functionalities include data pre-processing, clustering, classification, and regression. At their core, these tools employ various algorithms, such as artificial neural networks and Bayesian networks, for numerical data analysis and modelling. Graphical user interfaces facilitate data visualisation and algorithm parameter tuning. Additionally, some sophisticated or industry specific MLNA tools can directly query databases using SQL and might offer features automated machine learning.	Bertomeu et al. (2021) ; Bhattacharya et al. (2011) ; Etheridge et al. (2000) ; Green and Choi (1997) ; Koh and Tan (1999) ; Perols (2011)
CV	Video-based interview system; Drone-based inventory audit system	Computer vision empowers computers to interpret and act on visual data. Key properties include image and video acquisition for processing live or stored visual content. Pre-processing refines image quality, while feature extraction identifies crucial patterns. Object detection and classification enable recognition and labelling of visual elements. Depth estimation and 3D reconstruction provide spatial understanding.	Pentland et al. (2017) ; Christ et al. (2021)

Table 3
Categories and Properties of CAATTs.

Categories	Example artefacts	Properties	Exemplar research
GAS	ACL and CaseWare IDEA	Software packages with a database containing auditing rules, a database language that can interrogate the output of other information systems, and a user interface allowing the execution of standardised procedures.	Ahmi and Kent (2013); Braun and Davis (2003); Lin and Wang (2011)
CAS	SAPSECURE, CAMAP, and Bagheera-S modules	Software packages embedded into ERP or as an independent layer, with a database containing an instantiation of the control and monitoring process, a stream of metrics engineered out of key processes, a dynamic set of models to compare with the metrics, and dynamic exception metrics to trigger alarms.	Farkas and Hirsch (2016); Li et al. (2007) , Alles et al. (2004); Singh et al. (2013);

Table 4
Categories of Goal-Oriented at Three Levels of Actors.

Level	Goals	Explanation	Exemplar research
Individual	(1) Extrinsic goals	Individual practitioners are driven by extrinsic motivations, including job security, pursuing rewards and promotions, and avoiding negative outcomes such as reprimands or potential legal liabilities. Collectively, these motivations inspire them to utilise technologies that facilitate the achievement of employer's and professional's requirement.	Abou-El-Sood et al. (2015), Fischer (1996), Curtis and Payne (2014), Bierstaker et al. (2014)
	(2) Intrinsic goals	Individual practitioners are fuelled by intrinsic motivations, such as work enjoyment, autonomy in decision-making, intellectual stimulation, and continuous learning to remain at the forefront of their field. Motivated by these objectives, they seek to delegate mundane tasks to available technologies so that they can allocate more time to endeavours that align closely with their intrinsic motivations.	Bradford et al. (2020), Manita et al. (2020), Tiron-Tudor and Deliu (2022), Goto (2021, 2022)
Organisational	(3) Surveillance and control goals	Firms are motivated to establish stronger surveillance and control over business procedures, mainly for quality and risk control. They interact with CAATTs and AI artefacts to eliminate behaviours outside the narrowly defined "standard procedures." The goal also drives firms to ensure stronger senior surveillance over their teams, such as reviewing the team members' daily working performance.	Manson et al. (2001), Dowling and Leech (2014), Meredith et al. (2020)
	(4) Profit-oriented goals	The short-term business goal of every firm is to maximise its profitability by securing the revenue stream from clients and controlling the cost stream. To achieve the first objective, firms need to improve the input into the business process to ensure service quality. To achieve the second objective, firms need to optimise spending in their operating cycle. Firms must balance these two contradictory objectives to achieve the overarching profit-oriented goals.	Widuri et al. (2016), Siew et al. (2020), Manita et al. (2020)
	(5) Strategy-oriented goals	Firms are also motivated to sustain long-term business success. Rather than focusing on the current profitability, the long-term success is determined by its surviving strategy and future vision. For instance, amplifying the current revenue stream is a tactical goal, while diversifying the revenue streams is more about strategic orientation. This goal is usually realised by creating a distinctive and non-duplicable competitive advantage.	Krieger et al. (2021), Manson et al. (2001), Sarkar and Sriram (2001)
Professional association	(6) Legitimacy goals	A professional association is motivated to maintain the profession's social legitimacy. Externally, the professional association speaks for the profession against public criticism adversely affecting the profession. Internally, it establishes codes, standards and policies to ensure its existence meets other social members' expectations.	Dillard and Yuthas (2001), Alles et al. (2004), Munoko et al. (2020)
	(7) Sustainability goals	While maintaining legitimacy is current-oriented, this goal is more about the future. A professional association is driven by ensuring the profession can be sustainable and survive in the ever-changing environment. For instance, the professional association may reconfigure the role identify of the profession in the digital age to minimise the risk of being replaced by technology.	Goto (2022), Goto (2021), Tiron-Tudor and Deliu (2022)

some machine learning-based auditing systems developed by large accounting firms, such as Argus, are excluded from the list. This is because the existing body of literature offers limited discussion of these systems, primarily due to commercial licenses and intellectual property protection.

Subsequently, we identified two categories of CAATTs: generalised auditing software (GAS) and continuous auditing system (CAS) (Ahmi & Kent, 2013; Braun & Davis, 2003; Lin & Wang, 2011). This categorisation permits a more in-depth analysis of CAATTs with comparable properties and analogous functionalities. Generalised auditing software is auditing software with multiple data functions –

Table 5
Explanation of AI Affordances.

AI affordances Goals	Features required	Exemplars	Related Articles
Individual level: Make data-led decisions			
Quality-related goals (extrinsic and intrinsic). Externally, individuals strive to improve the quality of their work to gain recognition and rewards; Internally, they seek to boost self-esteem through the accomplishment of high-quality work.	Artefacts offer data insights into complex, judgement-intensive, and critical decision-making tasks, often beyond human capacity. Incorporating algorithmic insights can improve decision quality.	<i>"Because of deep learning's information identification function, the audit data warehouse covers semi-structured and unstructured data, describing the company from a variety of aspects and providing a wide range of audit evidence for decision-making" (Sun, 2019, p. 98).</i>	Santosuosso (2022), Pentland et al. (2017), Dong et al. (2018), Etheridge et al. (2000), Hajek and Henriques (2017)
Individual level: Allocate resources to higher level tasks			
Efficiency-related goals (extrinsic and intrinsic). Externally, individuals strive to enhance work efficiency to fulfill job requirements; Internally, they aim to make their jobs smarter, intellectually stimulating, and enjoyable.	Artefacts offer automation that can reduce human effort on mundane, repetitive tasks. Delegating those tasks to artefacts, humans can better utilise their expertise, judgment, and learning ability to tackle higher intellectual challenges.	<i>"Companies are finding that auditors are burdened by repetitive activities in audit processes and that RPA may help reduce this burden while simultaneously freeing up auditors to focus on value-enhancing tasks that require problem-solving, creativity, and human judgment" (Eulerich et al., 2022, p. 965).</i>	Ahmi and Kent (2012), Huang and Vasarhelyi (2019), Green and Choi (1997), Christ et al. (2021)
Organisational level: Standardise processes			
Controlling and supervising goals: firms aim to establish standardised procedures to minimise variation in practices and ensure their staff consistently follow these established protocols.	Artefacts organise data, information, and procedures into a structure, creating a path consistent with the prescribed methodology. This further offers standardisation of service and quality control of the output.	<i>"Provide an example of Deloitte's 'Cognitive Audit' strategy, which involves first standardising the audit process, after which the standardised processes are digitised. Then the digitised tasks are automated, followed by employing advanced analytics to the audit. Finally, cognitive (augmented) technology is used to transform the audit" (Munoko et al., 2020, p. 213).</i>	Krieger et al. (2021); Lin and Wang (2011), Sutton et al. (2016), Huang and Vasarhelyi (2019), Abdolmohammadi (1991), Manson et al. (2001)
Organisational level: Automate processes			
Optimise profit goals: firms aim to optimise profits by reducing costs and enhancing efficiency, allowing them to produce the output with lower-cost inputs.	Artefacts offer automation that can replace labour-based procedures with machine-based procedures, achieving better efficiency by reducing the cost of inputs and/or increasing the outputs.	<i>"RPA tools help businesses improve the efficiency of processes and the effectiveness of services. First, replacing the human workforce reduces the cost and processing time for high-frequency tasks. The running cost of RPA software is around one ninth that of employing a human being" (Huang & Vasarhelyi, 2019, p. 3).</i>	Zhang et al. (2021), Alles and Gray (2019), Chiu et al. (2014), Christ et al. (2021), Debreceeny and Gray (2011), Koskivaara (2004)
Organisational level: Boost workforce capability			
Optimise profit goals: firms aim to enhance their profitability by minimising expenses, particularly through labour cost reductions. Without sacrificing quality, they tend to assign more complex tasks to less expensive (and less experienced) staff.	Artefacts with decision and judgement support features enable less-experienced, less-skilled staff to perform more demanding and complex tasks. By bridging this gap, artefacts afford firms the opportunities to boost the capability of workforce and reduce labour cost.	<i>"Moreover, Big four companies have implemented chatbots that assist junior auditors in their tasks. Digital aid may also be offered to auditors. A chatbot may be enabled in audits to be used as a search tool for an online technical library ..." (Tiron-Tudor & Deliu, 2022, p. 266).</i>	Hansen and Messier Jr (1986), Fischer (1996), Fedyk et al. (2022), Goto (2022), Davis et al. (1997), Elliott (1986)
Organisational level: Diversify workforce			
Build competitive advantages: firms aim to establish competitive advantages by broadening their talent base, thereby reducing the dependence on professionals possessing 'irreplaceable skill sets.'	Artefacts capable of performing tasks to the same standard as professionals, while requiring IT technical skills rather than professional skills, present opportunities for firms to diversify their workforce.	<i>"This change contrasted with the more traditional view that auditors are isolated specialists in a homogeneous culture that lacks diversity.... [audit firms had only] an extremely narrow range of people, or one could call them very unilinear', as most auditors joined the firms... having experienced only the same CPA examination" (Goto, 2021, p. 96).</i>	Fedyk et al. (2022), Seethamraju and Hecimovic (2022), Munoko et al. (2020), Ahmi and Kent (2012), Krieger et al. (2021)
Organisational level: Develop new business models			
Build competitive advantages: firms aim to gain competitive advantages by transforming their existing business	Artefacts with adaptive and scalable features that enable the firms to develop a new business model, such	<i>"These technologies can drastically improve the audit quality and evolve the audit offer towards services with</i>	Chiu et al. (2014), Chen et al. (2017), Sun (2019), Charitou et al. (2004),

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Table 5 (continued)

AI affordances Goals	Features required	Exemplars	Related Articles
model and diversifying their revenue streams.	as changing a billing model from labour-driven to technology-driven, as well as offering new value-adding services.	<i>stronger added value as customers could correct these errors, prevent risks and continuously improve their systems"</i> (Manita et al., 2020, p. 6).	Alles and Gray (2019), Singh and Best (2015)
Professional association level: Claim legitimacy			
Preserving social legitimacy: professional associations seek to uphold their social standing by governing member behaviours and shaping societal perceptions of the profession.	Artefacts designed to enhance the quality and relevance of the profession's services empower professional associations to reinforce the profession's legitimacy within the broader community.	<i>"This led to enhanced professionalism with AI-empowered higher service quality (auditors reimagined as 'Data-empowered Advisor' and 'Analytics User'). These productivity and quality improvements enabled professional legitimacy to be maintained (auditors reimagined as 'Continuous Self-disruptor' and 'Guardian of Capitalism as before') and service opportunities to continue"</i> (Goto, 2021, p. 88).	Fedyk et al. (2022), Goto (2022), Calderon and Cheh (2002), Munoko et al. (2020), Sarkar and Sriram (2001), Manson et al. (2001), Alles et al. (2004)
Professional association level: Update knowledge base			
Sustainability goals: Professional associations aim to maintain their sustainability in an ever-changing environment, with a key strategy being the continual update of their knowledge base.	Artefacts featured with self-learning afford the profession an opportunity to extract latent knowledge from data, transforming previously uncodified patterns into new, actionable insights.	<i>"In contrast, a deep neural network receives raw input...and automatically learns features through model training. The layers in the hierarchical structure enable a deep neural network to identify data features without human intervention... Instead, the computer gathers this knowledge by learning from the raw image data"</i> (Sun, 2019, p. 93).	Goto (2021), Humpherys et al. (2011), Sutton et al. (2016), Tiron-Tudor and Deliu (2022)

Table 6
Mapping of Affordances among CAATTs and AI Artefacts.

Affordances	CAATTs GAS	CAS	Weak AI RPA	ES	Strong AI NLP	MLNA	CV
(1) Make data-led decision					•	•	•
(2) Allocate resources to higher level tasks	•		•		•	•	•
(3) Standardise processes	•	•	•	•			
(4) Automate processes	•	•	•		•	•	•
(5) Boost workforce capability	•	•		•	•	•	
(6) Diversify the workforce		•			•	•	•
(7) Develop new business models		•			•	•	
(8) Claim legitimacy			•		•	•	•
(9) Update knowledge base					•	•	•

such as display, reading, analysis, manipulation, sampling, and extraction (Bradford et al., 2020). ACL and CaseWare IDEA are the two most frequently discussed generalised auditing software packages, but other packages with similar functions are available, such as ProAudit, CCH, IRIS and Mersia (Ahmi & Kent, 2013). Most generalised auditing software can only perform some fundamental data analysis based on the output of clients' ERP systems. By contrast, a continuous auditing system allows auditors to conduct controls and substantive tests on transactions throughout the year to detect abnormal transactions and to ascertain whether any modifications have been made to the computer program (Li et al., 2007). A continuous auditing system examines the accounting process rather than the accounting output. Based on the system architecture, it can be either an embedded audit module in the ERP system or an independent monitor and control layer owned and operated by the auditors (Singh et al., 2013). These classic tools are distinct from strong AI because they adhere to specific rules and can only operate strictly within their pre-programmed functions. While they share similarities with weak AI in operating based on explicit rules, the key differences lie in their inability to adapt to new tasks and lack of engagement with practitioners' decision-making processes, areas where ES and RPA typically excel. Table 3 summarises and categorises the CAATTs discussed in the literature.

Understanding the properties of AI and CAATTs serves as a fundamental premise for examining the emergence of their respective affordances and constraints. It provides an avenue to understand why an artefact can or cannot be used in a certain way (Faraj & Azad, 2012). For example, auditors can use computer vision, rather than other machine learning tools, to detect untruthful information because there is a sensor capturing the interviewee's facial movement and rigidity. In the next section, we elaborated on the goal orientations of social actors when interacting with classic IT and AI artefacts, which is another fundamental premise for understanding the emergence of affordances (Leidner et al., 2018).

Goal orientation of Individuals, Firms, and professional associations

We analysed the social actors' goal orientation at three levels: individuals, firms, and professional associations. The multi-dimensional perspective aligns with prior affordance studies claiming that technology affordances vary among different societal levels, ranging from family to state (Faik et al., 2020). At the first level, individual actors refer to the practitioners using classic IT and AI artefacts in the working context. They are motivated by a series of extrinsic goals, such as securing jobs, obtaining rewards and avoiding punishment, as well as intrinsic goals such as making their job more intellectually stimulating and staying ahead of the curve.

At the second level, organisational actors are the firms that adopt and implement classic IT and AI artefacts. They are driven by the organisational goals to interact with the artefacts. The organisational goal is the state an organisation wishes to attain (Gagné, 2018). Although the existing literature acknowledges that organisational goals can be achieved through the collective action of individuals, it frequently fails to consider the potential discordance between individual goals and organisational goals (Burton-Jones & Volkoff, 2017). Because the organisation is a coalition of multiple stakeholders (including employees, owners, creditors, etc.), its goal cannot be interpreted simply as a collective goal of the technology's end users (employees). In fact, we find that the goal-orientation of a firm diverges from that of an individual user. For instance, firms adopt AI to optimise efficiency and reduce headcount, which is undesirable for employees but favourable for many other stakeholders (Krieger et al., 2021). Thus, we interpreted organisational goals as the ultimate state that the organisation aims to achieve, rather than a compilation of the goals of individual constituents (Gagné, 2018). In this regard, we found that firms are prompted to leverage classic IT and AI tools to impose control over individuals and work processes, optimise profit, and build competitive advantages.

At the third level, professional associations serve as a collective voice for individuals within a specific profession, facilitating social construction and negotiation of their professional identity and standards of practice (Greenwood et al., 2002). The overarching aim of such associations is to advocate and bolster the interests of the given profession, accomplished through a plethora of activities, such as establishing educational and training standards, formulating codes of ethics and conduct, and lobbying on behalf of the profession with governmental entities and other stakeholders. These associations further serve as dynamic platforms for members to connect, disseminate knowledge, exchange best practices, and remain informed about recent advancements within their respective fields (Cooper & Robson, 2006). Within the purview of our research, the professional associations are representative bodies of auditors or the broader accounting community, such as CPA. Our findings reveal that those professional associations are motivated to incorporate AI, and previously, CAATTs, into their agenda as a strategy to navigate crises, uphold social legitimacy, and ensure survival amidst a rapidly transforming environment. For instance, the Japanese Institute of Certified Public Accountants (JICPA), an accounting

Table 7
Explanation of Constraints.

AI Constraints Goals	Features imposing constraints	Exemplars	Related Articles
Individual level: Information bombardment Efficiency related goals: individuals aim to improve work efficiency to streamline tasks, making it simpler, smarter, and enjoyable.	Artefacts produce too much information, such as alarm floods or high false positive results, overwhelming actors and reducing efficiency.	<i>"If the number of alarms generated by the CMBPC system explodes, then it will hamper the ability of auditors and other enterprise personnel to react and correct the identified problems"</i> (Alles et al., 2006, p. 157).	Calderon and Cheh (2002), Bertomeu et al. (2021), Singh and Best (2015)
Individual level: Black-box decision making Quality-related goals: individuals aim to enhance the quality of work to receive job recognition and rewards, and to avoid punishment and liability.	Artefacts typically feature a black-box decision-making model such that actors do not understand how the artefacts reach a conclusion. This may undermine the decision quality.	<i>"However, the majority of systems designed to detect fraud reported by researchers aim to maximise the prediction accuracy, while disregarding how transparent they are. This factor has particular significance as the development of interpretable models is critical for supporting the investigation procedure in auditing"</i> (Craja et al., 2020, p. 4).	Koskivaara (2004), Santosuosso (2022), Hansen et al. (1992), S. Li et al. (2021), Perols (2011), Craja et al. (2020)
Individual level: Task displacement Job security-related goals: individuals aim to ensure stable employment and protect their roles against external disruptions or changing job requirements.	Artefacts capable of automating processes or supplanting professional knowledge may jeopardise practitioners' job security, as tasks become accessible to AI or individuals with diverse skills.	<i>"We must change our assumption of our human resources centred on people with accounting or financial backgrounds. Since data usage will be more important, we'll need more people with science backgrounds"</i> (Goto, 2022, p. 91).	Nelson et al. (2000), Fedyk et al. (2022), Alles and Gray (2019); Caster et al. (2021); Manson et al. (2001)
Firm level: Existing practice replication Build competitive advantages: firms aim to gain competitive advantages by transforming their existing business model and diversifying their revenue streams.	Artefacts are designed to replicate the existing practice (with better efficiency). Thus, firms are fixated on existing procedures, even if it may be necessary to deviate from established practices.	<i>"This is illustrated by the increased sharing of information facilitated by audit automation and consequently more intense reviewing by partners of work done by other audit staff. Another example whereby the domination structures within audit firms are reinforced and reproduced is the way in which increasing audit automation constrains the work behaviour of audit staff"</i> (Manson et al., 2001, p. 126).	White (1995), Hansen and Messier Jr (1986), Fischer (1996), Sutton et al. (2016)
Firm level: Responsibility gap generation Controlling and supervising goals: firms aim to establish standardised procedures to minimise variation in practices and ensure their staff consistently follow these established protocols.	Artefacts with complex and opaque structure may constrain the firm's ability to identify the causes of error and hold relevant people responsible for failures and mistakes (Failure could result from a design flaw or inappropriate use).	<i>"While the institution will be held accountable for AI, there may be no specific individual who could be held responsible for AI's failure, resulting in a responsibility gap probing developers of AI to enact safeguards against social sorting or grouping of people by AI"</i> (Tiron-Tudor & Deliu, 2022, p. 227).	Lehner et al. (2022); Munoko et al. (2020), Dillard and Yuthas (2001)
Firm level: Performance fluctuation Controlling and supervising goals: firms aim to establish standardised procedures to minimise variation in practices and ensure their staff consistently follow these established protocols.	Artefacts possessing unpredictable and uncontrollable performance attributes can constrain a firm's capacity to provide consistently high-quality services, as numerous factors may impact its performance.	<i>"Again, the SVM model achieves the best performance. The average testing accuracy, however, is only 56.17 percent, much lower than the model performance using the social media data. Note that the baseline model performance is lower than that of (previous studies) using the same financial ratios. This may have to do with treatment of missing values in the data sample"</i> (Dong et al., 2018, p. 476).	Hajek and Henriques (2017), Sun (2019), Etheridge et al. (2000), Krambia-Kapardis et al. (2010), Perols (2011)
Professional association level: De-professionalisation Sustainability goals: Professional associations strive to ensure their sustainability in a constantly evolving environment, ensuring their members consistently have employment opportunities.	Artefacts that can automate processes and discover patterns pose sustainability challenges for the entire profession. They could reduce the demand for traditional professional roles and dilute the exclusivity of expert knowledge.	<i>"They observe that the use of AI in place of a human is an enticing option since AI can work 24 h without breaks, and currently, there is no need for the employer to pay employee benefits and taxes for AI... at the extreme the replacement of auditors by data analytics</i>	Goto (2021, 2022), Seethamraju and Hecimovic (2022), Fedyk et al. (2022)

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Table 7 (continued)

AI Constraints Goals	Features imposing constraints	Exemplars	Related Articles
		firms, who have a different mindset than that of the traditional auditor" (Munoko et al., 2020, p. 15).	

professional body in Japan, undertakes numerous sense-making initiatives around AI with the primary aim of preserving the value intrinsic to the work of auditors and mitigating potential threats posed by the emergence of digital technology (Goto, 2022). Table 4 summarises and categorises the goal orientations of actors at three levels.

We identified the goal orientations of individual practitioners, firms, and professional associations within the auditing profession. Understanding how these goals intersect with the properties of CAATs and AI is essential to conceptualise affordances and constraints. Prior research has extensively examined goal orientation at the individual (Leidner et al., 2018; Leonardi, 2011) and organisational levels (Du et al., 2019; Strong et al., 2014), yet has limited discussion on professional associations.

Overview of AI affordances

AI affordances represent the goal-directed action potential with AI, a concept that is relational and a joint product of goal-oriented actors and the properties of AI tools (Faraj & Azad, 2012). Unlike explicitly definable goals and AI properties, affordances are often latent, meaning that they are not directly observable or inherently present without the context of actor-artefact interaction (Thapa & Sein, 2018). Instead, affordances can be 'retroduced' from the outcomes achieved by goal-directed actors utilising the artefacts, or even from the outcomes these actors aim to achieve (Bygstad et al., 2016; Volkoff & Strong, 2013). Therefore, when conceptualising AI affordances, we mapped the goal orientation of individual practitioners, firms, and professional associations, with the properties of five types of AI artefacts. We also referred to the literature that delineates the outcomes of using AI. The focus here is particularly on whether a specific AI tool enables an action that these actors can potentially take to accomplish their goal orientation. Table 5 presents the AI affordances through the interplay between the goal orientations of actors at three levels and the AI properties essential for affordance to emerge, as illustrated by examples in the literature and relevant studies.

In professional service environments, AI enables practitioners to make informed, data-driven decisions and reallocate resources towards more complex tasks. This is made possible by AI's superior data analysis abilities and its capacity to replicate human task performance. This technological empowerment serves two fundamental goals: achieving quality in complex, critical decision-making tasks for job recognition and penalty avoidance; and enhancing efficiency by automating mundane tasks so that practitioners to engage in intellectually stimulating activities and learning.

At the organisational level, firms are strategically oriented to establish and enforce standardised procedures to control operation, optimise profits through labour cost reduction, and build comparative advantages by diversifying talent pools and innovating business models. AI plays a vital role in supporting these objectives through its capabilities to organise information systematically, automate tasks that replicate professional knowledge, provide support and training for junior staff, and adapt to various business contexts. Consequently, AI facilitates a range of affordances for professional service firms, including standardising and automating of processes, enhancing and diversifying workforce capabilities, and innovating business models.

For professional associations, the integration of AI aligns closely with two critical objectives: maintaining their social legitimacy within the broader community and ensuring the profession's sustainability to the evolving environment. These goals are supported by AI's capacity to develop data insights that were previously unpossessed by the profession, and to elevate the quality and relevance of professional services through sophisticated data analysis. With these capabilities, AI affords professional associations an opportunity to claim legitimacy and update their knowledge base.

We also repeated the coding process for both generalised auditing software and continuous auditing systems. This allowed us to contrast affordances between AI and CAATs. The findings from this process, which provide a clearer delineation between AI and classic IT in terms of their respective affordances, are presented in Table 6.

A close examination reveals that certain affordances are exclusive to AI, particularly to more advanced AI tools, such as facilitating

Table 8
Mapping of Constraints to CAATs and AI Artefacts.

Constraints	CAATs	Weak AI	Strong AI				
	GAS	CAS	RPA	EDS	NLP	MLNA	CV
(1) Scope limitation	•	•					
(2) Information bombardment		•			•	•	
(3) Task displacement			•	•	•	•	•
(3) Black-box decision making					•	•	•
(4) Existing practice replication	•	•	•	•			
(5) Responsibility gap generation					•	•	•
(6) Performance fluctuation					•	•	•
(7) Dependence on clients' systems	•	•					
(8) De-professionalisation			•	•	•	•	•

Table 9
Integrated Affordance and Constraints Framework.

Goals	Property	Affordances	Constraints	Potential outcomes
Individual practitioners				
Quality	Rule – discovering	Data led decision	Black-box decision	Quality and transparency trade-off
Efficiency	Both Rule – discovering	Re-allocate resource	Information bombardment	Role shift from executor to monitor and investigator
Job Security	Both		Task displacement	Higher risk of job loss if skills do not evolve
Professional service firms				
Control and surveillance	Rule – based Rule – discovering	Standardise practice Responsibility gap generation & performance fluctuation		Stronger control in low-judgemental tasks and weaker control in high – judgemental tasks
Profit-oriented	Both	Automate processes & boost workforce capability		Higher profit from cost reduction, efficiency improvements, and enhanced capabilities
Strategy-oriented	Rule – based Rule – discovering	Diversify the workforce & develop new business models	Existing practice replication	Transfer from knowledge-intensive practice to technology-intensive practice
Professional associations				
Legitimacy	Both	Claim legitimacy		Enhanced legitimacy from enhanced quality and relevance
Sustainability	Rule – discovering Both	Update knowledge base	De-professionalisation	Profession transformation from knowledge occupiers to knowledge co-producers

data-led decision-making for individual practitioners and creating new knowledge for professional associations. Essentially, these actions become near impossible, or at least highly challenging, without the use of the relevant advanced AI tools. Concurrently, it's worth noting that some affordances are shared across CAATs, weak AI, and strong AI artefacts. For instance, firms can automate business processes with the following artefacts: robotic process automation to replace labour in configurable and repetitive tasks (Eulerich et al., 2022); generalised auditing software, which “involves simple automation of manual procedures” (Singh et al., 2013, p. 306); continuous auditing systems which “would predictably follow the path of first automating existing manual audit procedures” (Alles et al., 2006, p. 138); computer vision which “identified facial rigidity in a scientifically controlled automated interview as a possible alternative” (Pentland et al., 2017, p. 991); and natural language processors that conduct an automated text search of management disclosure for material misrepresentations or omissions that are seemingly intentional” (Brown et al., 2020, p. 240).

Overall, the affordances of classic IT tools are largely replicated by AI artefacts, but AI, particularly advanced AI, provides unique affordances that unlock actions and outcomes that would otherwise be unattainable. This comparative assessment provides a foundational basis for our discussions on the strategic implications of AI for individual practitioners, firms, and professional associations.

Overview of AI constraints

Like affordances, technological constraints are also relational, wherein the specific properties of a technology could potentially inhibit actors' potential for action (Majchrzak & Markus, 2012). These constraints may surface when actors struggle to use technology to take desirable actions for achieving their goals (Leonardi, 2011; Leonardi & Vaast, 2017). When these constraints emerge, actors are inclined to either modify the technology or adopt alternative technologies for goal fulfilment (Bagozzi, 2007; Leonardi, 2011). To conceptualise constraints posed by AI, we iteratively examined the properties of the five AI artefacts, alongside a careful consideration of the potential negative outcomes arising from their use. Our focus was directed towards identifying the actions that were hindered by AI use, thereby giving rise to these adverse outcomes. Through this approach, we identified multiple constraints imposed by AI on actors at individual, organisational, and professional association levels, hindering their pursuit of goals.

First, for individual practitioners, some AI artefacts tend to generate an overwhelming amount of information, posing challenges to efficiency-related goals by inducing decision fatigue and diminishing work efficiency. Simultaneously, the opaque nature of AI applications, especially those built on neural network, obstructs quality-related goals, compromising the quality and transparency decision-making. AI also makes it harder for practitioners secure their positions, as it increasingly takes on tasks with superior quality and efficiency. Second, for professional service firms aspiring to innovate and secure competitive advantages, the replication of existing practices by certain AI artefacts emerges as a significant hindrance. While these systems might enforce procedural compliance, they inadvertently stifle innovation and flexibility. Moreover, the complex structure of some AI solutions can create a responsibility gap and lead to inconsistencies in performance, thereby obstructing effective oversight and quality control of work processes. Third, professional associations striving to maintain social legitimacy are confronted with task displacement due to AI automation,

challenging the profession's relevance and diminishing associations' effort to claim legitimacy. As shown in Table 7, these constraints arise when specific features of AI artefacts intersect with goal orientations, leading to unintended barriers in achieving desired outcomes.

We extended our analysis to generalised auditing software and continuous auditing systems, enabling us to differentiate the constraints associated with AI and CAATs. It is worth noting that two unique constraints – 'scope limitation' and 'dependence on clients' system' – are discernible in CAATs but absent from AI artefacts. Generalised auditing software and continuous auditing systems are customarily used to process structured numerical and categorical data existing in clients' systems. However, they encounter limitations when tasked with processing other valuable forms of data, such as contracts, management disclosures, images, and meeting records. This 'scope limitation' significantly curtails the range of information practitioners can access (Singh & Best, 2015). This limitation could consequently confine practitioners' decision-making processes and may negatively impact their work performance.

Moving to the firm level, the constraint termed as 'dependence on clients' system' is again critical. Generalised auditing software and continuous auditing systems exhibit high dependence on the client's existing information systems. If these systems are outdated, complex, or lack interoperability, firms may face challenges delivering their services effectively. They might resort to manual procedures to compensate for these system inadequacies, which could significantly escalate operational costs and reduce efficiency (Vasarhelyi and Romero, 2014; Alles et al., 2006). In light of these distinct constraints presented by CAATs, we have compiled our comparative analysis of the constraints imposed by CAATs and AI, which is presented in Table 8.

As observed through our table, an interesting pattern has surfaced. Besides the two unique constraints pertinent to CAATs that we previously discussed, CAATs and weak AI share several common constraints, such as the 'replication of existing practices.' Strong AI, on the other hand, exhibits more distinctive constraints, largely because of their common properties.

For example, due to the complex nature of their underlying algorithms, machine learning number analytics impose 'black-box decision making', where "most of the fraud detection systems reported in the literature have been developed by maximising only the accuracy of prediction, while neglecting the interpretability" (Hajek & Henriques, 2017, p. 140). The same property and constraint is found in computer vision, which "is powered by algorithms that are too complex to be understood and interpreted" (Sun, 2019, p. 90), and natural language processing, "despite their prediction accuracy, disregarding how transparent they are" (Craja et al., 2020, p. 4). Hence, it is safe to conclude that 'black-box decision making' is a shared constraint among strong AI artefacts, but it doesn't feature in the realm of weak AI or CAATs. Other unique constraints imposed by strong AI include the 'generation of responsibility gaps' and 'performance fluctuation'. Recognising these distinctions in constraints between classic IT and AI, and even within the AI category between strong and weak AI, leads us to further discuss its implications.

An integrated framework

In this section, we consolidate the findings to present an integrated framework that discusses the affordances and constraints of AI within professional services. AI affordances and constraints emerge from the interplay between the goal orientation of three levels of actors and the inherent properties of AI technology (Burton-Jones & Volkoff, 2017; Strong et al., 2014). While AI artefacts possess a range of properties – including interface design, arrangement, and architectural structure (Faraj & Azad, 2012) – our focus is primarily on their core properties: rule-discovering and rule-based. These properties are fundamental to the emergence of related affordances and constraints. As we delve into how these affordances and constraints interact, we begin to articulate the potential outcomes AI brings to individual practitioners, professional service firms, and professional associations. Table 9 illustrates the relationships between the elements we covered, presenting as an integrated framework.

At the individual level, practitioners are motivated to employ AI by a spectrum of intrinsic and extrinsic goals (Deci et al., 2017), which can be broadly categorised into objectives related to quality, efficiency, and job security. Among these, quality-related goals align with the rule-discovering properties of AI, enabling practitioners to make data-led decisions, rather than relying on intuition, heuristics, and experience (Sun, 2019). This promises an enhancement in the quality of decision-making. However, the very features that facilitate this advanced decision-making, such as the use of neural networks, can lead to black-box decisions (Seethamraju & Hecimovic, 2022). This opacity potentially undermines the transparency and accountability of the decision-making process, introducing a dilemma for practitioners: the choice between improved decision quality and the maintenance of decision-making transparency. Therefore, practitioners may need to carefully assess the trade-off between quality and transparency in various decision-making contexts.

Furthermore, practitioners are driven by the desire to enhance the efficiency of their work, prompting them to delegate mundane, repetitive tasks to machines. This strategy allows them to allocate their time and effort more effectively toward tasks requiring critical judgment and intellectually stimulating (Eulerich et al., 2022). Both rule-based and rule-discovering AI technologies facilitate this shift by replicating human activities. However, the rule-discovering AI, in particular, might frequently uncover unexpected findings and trigger alarms, leading to an information overload (Bertomeu et al., 2021). This barrage of alerts can inadvertently compromise the practitioners' efficiency, as considerable time must be devoted to addressing them. The integration of both types of AI, nonetheless, holds the potential to transform practitioners' roles from task executors to AI monitors and investigators. Instead of manually performing tasks themselves, practitioners would engage with the AI's output, investigating exceptions and unusual items to bridge any gaps.

Finally, while leveraging AI to enhance work quality and efficiency facilitates job recognition and security, AI introduces a notable constraint from a different perspective: it diminishes the number of tasks available to practitioners (Fedyk et al., 2022; Munoko et al., 2020). This task reduction is primarily due to AI's capacity to both automate processes with higher quality, efficiency, and to do so at

Table 10
Across Level Goal Alignment and Conflict.

Goal alignment and conflict	Practitioners	Firms	Professional	Control	Profit	Strategy	Legitimacy	Sustainability
Practitioners	Quality Quality				✓			✓
	Efficiency					✓		✓
	Security					×	×	
Firms	Control	✓						
	Profit		✓	×				×
	Strategy			×				×
Associations	Legitimacy	✓	✓					
	Sustainability					×	×	

✓ Goal alignment × Goal conflict.

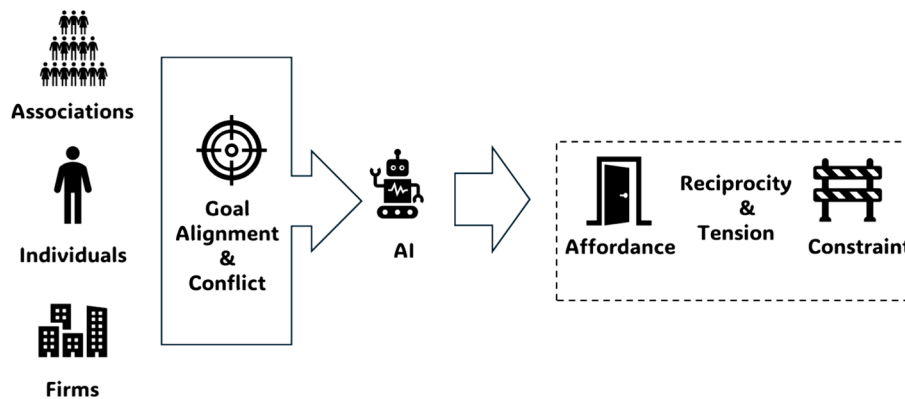


Fig. 1. Reciprocity and Tensions between Affordances and Constraints across Levels.

lower cost. Initially, this automation targeted low-value, mundane tasks, but it is increasingly encroaching on areas requiring high judgment, where practitioners previously held exclusive expertise. As AI continues to evolve and take on these complex tasks, practitioners face a pivotal challenge: without acquiring new skills and proactively adapting to shifts in their roles, it is plausible to anticipate a future where the tasks traditionally performed by practitioners become increasingly scarce.

For professional service firms, the motivation to integrate AI stems from multiple objectives: enhancing control and surveillance over practitioners and their work; optimising profit through revenue augmentation and cost reduction; and forging strategic advantages by transforming their business models (Krieger et al., 2021; Manson et al., 2001). When focusing on control and surveillance, AI's varied properties introduce distinct affordances and constraints. Rule-based AI, for example, facilitates the attainment of this goal by standardising practices. This standardisation is achieved by implementing consistent, predefined rules that guide practitioners' actions, ensuring uniformity in the quality of work and adherence to firm policies. Conversely, rule-discovering AI poses challenges to maintaining control and surveillance. This form of AI learns and applies rules derived from data analysis, which means the quality of work is heavily influenced by the quality of the underlying data, rather than the performance of individual practitioners (Munoko et al., 2020). This reliance on data quality can lead to fluctuations in performance and introduces a gap in responsibility, where failures are difficult to attribute to specific practitioners. Thus, implementing both types of AI presents a complex picture for firms. While rule-based AI can ensure stronger control and consistency in tasks requiring low judgment, the use of rule-discovering AI might result in less control and inconsistency in outcomes, unless firms establish clear policies to manage this unique challenge.

Firms' profit-oriented goals find a synergy with the capabilities of both types of AI. These AI artefacts excel in automating processes that were once manual, significantly enhancing efficiency with lower labour costs. Moreover, data insights and guidance from AI empower the workforce, especially junior staff, to undertake tasks that traditionally demanded a higher level of experience and expertise (Mascha & Smedley, 2007; Tiron-Tudor & Deliu, 2022). This empowerment is particularly cost-effective, as firms often incur much lower labour costs for junior staff compared to their seniors. Consequently, the integration of AI into business operations promises substantial financial benefits. At the strategic level, AI presents a twofold dynamic that leans more towards affording opportunities than imposing constraints. On one hand, rule-based AI, with its focus on automation and enhancing efficiency, tends to replicate existing practices, which could potentially reinforce the status quo and create obstacles for transformative change (Manson et al., 2001). On the other hand, rule-discovering AI affords firms the opportunity to diversify their workforce and pioneer new business models. With strong data analysis and information processing capabilities, rule-discovering AI reduces firms' dependency on practitioners. This shift allows firms to restructure their workforce, incorporating new roles such as data scientists and AI specialists (Goto, 2022). Implementing AI also enables the exploration of novel business opportunities. For instance, firms can transition from offering traditional auditing services to providing value-added data insights to clients (Seethamraju & Hecimovic, 2022). This

Table 11
Reciprocity and Tensions between Affordances and Constraints across Levels.

Relationships	Goal alignment	Reciprocity between affordances
R1	Quality (IP) & Control (PSF)	Make data-led decision (IP) & Standardise process (PSF)
R2	Quality (IP) & Legitimacy (PA)	Make data-led decision (IP) & Claim legitimacy (PA)
R3	Efficiency (IP) & Profit (PSF)	Re-allocate resource (IP) & Boost workforce capacity (PSF)
R4	Efficiency (IP) & Legitimacy (PA)	Re-allocate resource (IP) & Claim legitimacy (PA)
	Goal conflict	Tension between affordances and constraints
R5	Profit (PSF) vs. Security (IP)	Automate process (PSF) vs Task displacement (IP)
R6	Strategy (PSF) vs. Security (IP)	New business model & Diversify workforce (PSF) vs Task displacement (IP)
R7	Profit (PSF) vs. Sustainability (PA)	Automate process (PSF) vs De-professionalisation (PA)
R8	Strategy (PSF) vs. Sustainability (PA)	New business model & Diversify workforce (PSF) vs. De-professionalisation (PA)

IP=individual practitioners; PSF=professional service firms; PA=professional associations.

evolution signifies a strategic shift from knowledge-intensive to technology-intensive, redefining the firm's core competencies and the value proposition it offers to its clients.

For professional associations, navigating the AI era entails two fundamental goals: maintaining social legitimacy and ensuring the long-term sustainability of the profession (Goto, 2021, 2022). The pursuit of legitimacy finds a harmonious alignment with the capabilities of both types of AI. By incorporating AI, the auditing profession can boost the quality and efficiency of services. This enhancement enables professional associations to persuasively demonstrate to clients, regulators, and public that, through the integration of AI, auditing services are delivering superior quality and contributing value to society by utilising resources more efficiently.

However, in the pursuit of sustainability goals, AI acts as a double-edged sword for professional associations. On the one hand, rule-discovering AI, with its capacity to uncover new knowledge from data, offers these associations the opportunity to refresh and expand their knowledge bases, thereby enhancing the profession's adaptability to the new environment. On the other hand, the introduction of both types of AI has a profound effect on diminishing the significance of the traditional roles played by auditing profession, leading to a phenomenon known as de-professionalisation (Munoko et al., 2020). This occurs as AI technologies take over tasks that were previously the exclusive domain of highly skilled professionals, thereby reducing the demand for traditional auditing skills and knowledge. In response to these transformative pressures, it is reasonable to predict that professional associations may steer the auditing profession from being mere occupiers of specialised knowledge to becoming co-producers of knowledge alongside AI. For instance, the auditing profession could serve as a guardian of the knowledge generated by AI, ensuring it is utilised in ways that align with society's best interests. By embracing this role, professional associations can ensure the profession remains dynamic, relevant, and sustainable.

Reciprocity and tensions between affordances and constraints across levels

In the above sections, our analysis focused on how affordances and constraints emerge from the interplay between actors' goals and AI properties at a detailed, 'zoomed-in' level. We examined those elements distinctly and separately at the levels of individuals, firms, and professional associations. Upon expanding our view to a broader, 'zoomed-out' perspective, we discovered the presence of goal conflicts and alignments across these various levels of actors. We then delved into exploring how these alignments and conflicts might influence the affordances and constraints across different strata.

As illustrated in Table 10, there are four alignments and four conflicts of goals among three levels. Driven by a combination of extrinsic and intrinsic goals, individual practitioners are motivated to enhance the quality and efficiency of their work. This pursuit resonates closely with the overarching aim of professional associations to maintain the legitimacy of the profession as a whole. The alignment between individual and collective goals is rooted in a mutual commitment to uphold ideal professional values and the integration of these values with managerial and social demands (Abbott, 1988; Greenwood et al., 2002). By advocating for the preservation and recognition of the intrinsic value of professional services, particularly efficient use of resources and high-quality outputs, professional associations strive to ensure that their professions continue to be esteemed within the social division of labour (Tidmarsh, 2022). The efficiency and quality goals of practitioners are also aligned with the objectives of professional service firms, which prioritise control and profit. Firms place a significant emphasis on control as a means to ensure the consistency and quality of their services (Manson et al., 2001; Seethamraju & Hecimovic, 2022). This focus on control parallels an individual practitioner's pursuit of quality, as both aim to enhance the value delivered to clients. Similarly, firms' objective of optimising profit finds a natural alignment with practitioners' efficiency; when practitioners' working efficiency improves, firms not only streamline operations but also reduce costs, thereby enhancing profit margins (Oral & Yolalan, 1990). This synergy between practitioners' goals of efficiency and quality and firms' aims of control and profit underlines a mutually beneficial relationship. It ensures that while practitioners are focused on delivering high-calibre work efficiently, firms can maintain high standards of service and achieve their financial objectives.

Nevertheless, the profit maximisation and strategic orientation of firms inherently conflict with individual practitioners' needs for job security and professional associations' goal of sustainability. Optimising profit often entails delivering services with cheaper inputs, which could mean replacing professional practitioners with machines or other labourers if they prove to be more cost-effective (Huang & Rust, 2018). Such practices, while beneficial for the firm's bottom line, jeopardise the job security of practitioners. Furthermore, firms' strategic orientation, which seeks to explore new business models and diversify workforces, aims to mitigate risks associated with reliance on a single revenue stream or talent pool (Frumkin & Keating, 2011). Although prudent from a business perspective, this strategic diversification can create instability for practitioners accustomed to traditional roles and practices.

Professional associations, which advocate for the profession's sustainability, also view these trends with concern, as they can lead to a devaluation of specialised skills and undermine the long-term viability of the profession (Greenwood et al., 2002). Thus, while firms may pursue these strategies to safeguard their future and maximise profitability, they inadvertently conflict with the best interest of practitioners and professional associations.

When interacting with AI properties, goal alignment and conflict dynamics manifest as reciprocal and tensional relationships between affordances and constraints across different levels of actors. Reciprocity in this context implies that the actualisation of one affordance can facilitate the actualisation of another affordance at a different level, creating a symbiotic enhancement of capabilities and opportunities across the spectrum of actors involved (Frey & Grafton, 2014; Renno, 2017). On the other hand, tension arises when the actualisation of an affordance either complicates the actualisation of another affordance at different levels or imposes constraints on actors at different levels in pursuing their goals (Chan et al., 2019; Gibbs et al., 2013; Haag et al., 2022). This tension can lead to a competitive or adversarial dynamic, where the gains for actors at one level come at the expense of actors at another level. This complex interplay of reciprocity and tension, highlighting the intertwined and sometimes conflicting pathways through which AI influences the pursuit of goals across different levels of actors, is illustrated in Fig. 1 below.

Table 11 below provides a summary of the reciprocity and tensions between affordances and constraints across the three levels of actors in our research context.

As mentioned earlier, the quality-driven goals at the individual level are in harmony with the control goals at the firm level and the legitimacy goals at the professional association level. This congruence fosters a reciprocal relationship among the affordances at different levels: individuals to make data-led decisions, professional associations to claim legitimacy, and firms to structuralise their processes. When practitioners utilise AI to inform their decision-making, professional service firms can standardise and structuralise processes. This is largely because AI enhances consistency in outcomes and mitigates personal biases, among other factors, leading to more reliable and predictable workflows (R1). Simultaneously, as practitioners harness AI to elevate the quality of their work, the improved outcomes contribute to a stronger foundation for professional associations to assert the relevance and value of the profession. Enhanced decision quality, underpinned by data-driven insights, showcases the profession's commitment to excellence. This, in turn, makes it easier for professional associations to advocate for the importance and societal value of professional services, reinforcing their legitimacy (R2).

In addition, practitioners' pursuit of efficiency dovetails with firms' profit-oriented objectives and professional associations' goals for legitimacy. This alignment translates into a mutual reinforcement of reallocating time and effort, boosting workforce capacity, and claiming legitimacy. Practitioners can reallocate their efforts towards more complex, high-value activities by employing AI to automate routine tasks. This reallocation directly ties into AI's affordance to boost firms' workforce capacity. As practitioners engage in more sophisticated, high-value work, the collective capability of the firm's workforce is elevated (R3). For professional associations, the shift of practitioners towards higher-level tasks facilitates an easier claim to legitimacy. This shift is critical as it aligns the profession with roles of greater social importance. Moreover, the resultant efficiency highlights enhanced use of resources and improves societal outcomes, reinforcing the profession's value and relevance (R4).

Moving to goal conflicts, our findings highlight a tension between the profit and strategic orientations of firms on one hand, and individuals' goal of job security and professional associations' aim of sustainability on the other. Focusing on the tension between individual practitioners and firms, two key areas stand out. First, firms are increasingly using AI to automate tasks, leading to the displacement of traditional practitioner roles. As AI systems take over routine tasks with greater efficiency and accuracy, practitioners are forced to relinquish these responsibilities. This shift creates a conflict, as individuals face job insecurity and the potential erosion of their professional roles and identities (R5). Second, in their quest for innovation and competitiveness, firms are adopting AI along with new business models and diversified workforce. This shift involves hiring experts from multiple backgrounds, such as data scientists and AI specialists, who bring a different set of skills to the table. These new business models no longer rely solely on practitioners' knowledge and expertise. While this diversification can lead to more innovative and adaptive business strategies, it also results in the displacement of traditional practitioners (R6).

Firm's shift towards automation and diversification also poses a profound challenge for professional associations. The tension can be observed in two main areas. First, firms' AI-enabled automation constrains the profession's sustainability by diminishing the demand for specialised human expertise. As AI takes over more tasks, fewer tasks remain for professionals, making it increasingly challenging for professional associations to justify the continued value of their members to society (R7). Moreover, diversified workforce and business model dilute the profession's significance, as job content that once required exclusive skills and knowledge becomes accessible to, and executable by, a broader range of professionals and technologies. This democratisation of skill and knowledge undermines the exclusivity of the profession, thereby reducing its perceived value (R8). AI's prevalence prompts a re-evaluation of what constitutes professional expertise and how it is valued in the marketplace, pushing professional associations to navigate a landscape where traditional roles are being redefined and the boundaries of professional jurisdiction are in flux.

Discussion

Our research contributes to affordance theory by introducing a novel conceptualisation on reciprocity and tension between affordances and constraints. It also has relevance to the study of AI within professional service sectors, laying out a research agenda for future inquiries. Moreover, although our findings were primarily based on the auditing industry, their practical implications are broadly applicable across various professional services due to common characteristics (Sampson, 2021; Spring et al., 2022). This suggests that our insights could provide strategic value to practitioners, professional service firms, and professional associations alike, offering guidance in navigating the complexities introduced by AI.

Theoretical Implications for Affordance Literature.

The current advancement in affordance theory, particularly its application across a spectrum of technological issues, has demonstrated its efficacy and robust analytical power. A significant portion of literature within this domain utilises affordance theory to explore the organisational transformations prompted by IS artefacts, highlighting the process of affordance actualisation (Haag et al., 2022; Nevo et al., 2021). This process describes how individuals within an organisation engage in goal-directed actions with IS artefacts that lead to tangible outcomes at the individual level, collectively contributing to attaining organisational objectives (Sampson, 2021; Spring et al., 2022). Such studies offer valuable insights, yet they frequently rest on the contentious assumption that the goals of individuals and organisations are inherently aligned. However, recent scholarly attention increasingly recognises the existence of goal conflicts and tensions between different affordances, challenging this assumption and suggesting a more complex interplay between individual and organisational actors (Haag et al., 2022; Nevo et al., 2021). For instance, Haag et al. (2022) theorised the notion of deviant affordance, where “action possibilities that arise from a relation between a goal-oriented user and technological properties and whose actualisation the organisational IT policies specify as undesired (p. 2113)”. This notion is similarly explored under the term ‘unintended affordances’ (Burke & Wolf, 2021; Vollenberg et al., 2023), where IS artefacts provide individual actors with opportunities to engage in activities that conflict with the organisation’s objectives in implementing such technology.

Our findings unveil a nuanced perspective that parallels and diverges from the established notion of deviant affordance. On the one hand, we observed reciprocal relationships between AI affordances across various actor levels, highlighting how the actualisation of affordances on the actor level can enable or facilitate the actualisation of affordances on other levels. For instance, AI affords individuals the chance to make data-led decisions, firms to standardise processes, and professional associations to assert legitimacy, exemplifying such reciprocity. This reciprocal relationship has been well explored in existing literature discussing affordance strands or affordance networks (Abouzahra & Ghasemaghaei, 2022; Burton-Jones & Volkoff, 2017; Leidner et al., 2018). On the other hand, our findings also uncover a tensional dynamic among AI affordances at different levels, where the actualisation of affordances by one level’s actor places constraints on other levels’ actors in achieving their objectives. This tension is exemplified when firms leverage AI to automate processes, constraining individual practitioners’ ability to secure their jobs and challenging professional associations’ efforts to sustain their legitimacy. The roots of these reciprocal and tensional relationships lie in the alignment and conflict of goals among the actors. Our analysis, as illustrated in Fig. 1, reveals that when interacting with AI’s properties, goal alignment and conflict develop as reciprocal and tensional relationships between affordances and constraints across different actor levels.

Our analysis introduces an expansion to the current understanding of deviant affordance. While IS artefacts afford individual actors an opportunity to undertake activities that may contradict the organisational objectives behind implementing such technology (Haag et al., 2022), it is equally true that IS artefacts afford organisational actors the possibility of undertaking activities at odds with individuals’ personal or collective goals. This observation compels us to propose a broader conceptualisation of deviant affordance, framing it as *the inherent tension within the affordances provided by IS artefacts to actors—be they individuals or organisations—whose goals conflict; under such tension, the actualisation of affordance by one actor imposes constraints on others*. This conceptualisation emphasises that deviant affordances could exist in both directions, rather than being singularly attributed to an individual’s deviant use of IS artefacts leading to undesired consequences for organisations. It invites a more profound exploration of how IS technologies are used and managed, broadening our understanding of the potential conflicts and constraints within and across sociotechnical relationships (Asatiani et al., 2021; Hund et al., 2021). This approach also underscores the importance of anticipating and mitigating adverse impacts that may arise from the divergent goals of different actors.

Implications for Research on AI in Professional Services and Future Research Opportunities.

Our exploration into AI affordances and constraints across various levels of actors within a professional services sector illuminates both the transformative potential and the challenges of AI integration. It proposes two affordance-based frameworks and outlines potential relationships for further investigation. The first framework, outlined in Table 9, illustrates the relationships among actors’ goals, AI properties, affordances and constraints, and potential outcomes. The second framework, detailed in Table 11 and Fig. 1, explores the reciprocity and tensions between affordances and constraints at various levels. These frameworks could serve as a cornerstone for future research into the interactions between AI and individual practitioners, professional service firms, professional associations, and broader themes within the professional services sector.

For researchers focused on **individual practitioners**, we propose the dual impact of AI, especially those applications with rule-discovering properties, that can enhance decision making quality but reduce decision transparency. The existing literature abundantly addresses the technological aspects of explainable AI (Adadi & Berrada, 2018; Barredo Arrieta et al., 2020), yet there has been less focus on strategies human actors could employ to navigate situations where AI’s transparency is limited. Key research questions to consider include:

- How do practitioners develop and implement strategies to maintain their decision-making transparency in scenarios where AI transparency is limited?
- How do practitioners manage the trade-off between transparency and quality in the context of data-driven decision-making?

Our findings indicate that AI has the potential to transform the role of practitioners from task executors to monitors and investigators of AI outcomes. This transformation underscores the dynamics of task delegation and supervision within AI-enhanced environments. While current IS literature offers a comprehensive framework for task delegation in modern technological contexts (Baird & Maruping, 2021), and studies like Fügner et al. (2022) and Song and Lin (2023) explore AI delegation in laypeople’s daily tasks, the unique characteristics of professional services suggest that there are still unexplored areas (Spring et al., 2022). Future research could delve into:

- The effect of practitioners' intervention on AI-driven outcomes: Does their intervention improve or undermine the decision quality?
- The best practices for task delegation with AI: How can we leverage the combined strengths of practitioners and AI for decision quality and efficiency?

For studies focused on the **firm level**, our research uncovers the twofold impact of AI on professional service firms' control and surveillance over practices and practitioners. AI affords firms stronger control over low- judgemental tasks, but it may diminish firms' control over complex, high- judgemental tasks. This dichotomy warrants future research. Echoing the notions of related research (Ahmed et al., 2022; Monod et al., 2024), our study calls for an in-depth investigation into how firms can strike a balance between empowering practitioners with AI and preserving managerial control, while also promoting responsible and accountable decision-making with AI:

- What governance structures and policies best promote accountable decision-making with AI in professional services firms?

We also contend that AI holds strategic implications for professional services firms, potentially transitioning them from knowledge-intensive to technology-intensive practices. While existing literature predominantly explores AI adoption, implementation, and short-term impacts within professional service firms (Spring et al., 2022; Trocin et al., 2021; Yang et al., 2024), it notably lacks an analysis of long-term ramifications. Consequently, research that delves into the transformative effects of AI on professional services firms, particularly through longitudinal studies, stands to offer substantial value:

- How does AI transform professional services firms in the long term, specifically in terms of their operational model, value proposition, and competitive landscape?

Future studies can also focus on **professional associations**, and the fact that de-professionalisation poses significant threats to the sustainability of entire professions. The introduction of AI, unlike traditional technologies, brings an unprecedented crisis due to the displacement of an increasing number of tasks previously carried out by professional services (Acemoglu et al., 2020; Frey & Osborne, 2017), requiring substantial changes within the profession. Extant literature has underscored the pivotal role of professional associations in institutionalising changes to manage crises effectively (Greenwood et al., 2002). However, current studies have not extensively explored this topic. Individual practitioners may find themselves ill-equipped to tackle this *meta*-crisis alone, yet the collective might they wield through professional associations often proves sufficient. Thus, it would be particularly intriguing to examine how professional associations manage the crisis. Future research could delve into:

- How do professional associations deploy strategies and initiatives to manage crisis and sustain the viability of professions in an AI disruptive environment?

Within the **broader context of professional services**, our study underscores the reciprocal and tensional relationship between AI's affordances and constraints across various levels of actors. This is an underexplored area, as previous research, with just a few exceptions, has often been siloed and focused on a single level of actors (Faulconbridge et al., 2023; Goto, 2023). We therefore recommend a cross-level analysis of AI impact to bridge this gap. Future studies could adopt a more holistic perspective, collecting empirical evidence on how AI's affordances and constraints interact between practitioners and firms. A potential research question to guide this inquiry might be:

- How do the affordances and constraints of AI interact across different levels of actors within professional services, and what are the outcomes of such interactions?

Finally, investigating how the affordances and constraints of AI manifest in different professional service industries (e.g., law, healthcare, consulting) could reveal industry-specific dynamics and yield cross-industry insights. It is intriguing to observe that despite the common characteristics shared by professional service sectors (Sampson, 2021; von Nordenflycht, 2010), some studies offer vastly differing predictions regarding AI's impact, with the risk of AI displacement estimated at 94 % for auditing and a mere 3.5 % for legal services (Frey & Osborne, 2017). Thus, comparative studies can enrich our understanding of the nuanced differences within sectors of professional services. Studies like Spring et al. (2022), which draw comparisons within professional services between law firms and accounting firms, should be highly valued.

Practical Implications for Individual Practitioners: Make Decisions with AI.

As we delved into AI affordances and constraints for individual practitioners, a recurring theme became evident: decision making. Unlike classic IT, AI can be regarded as agentic artefacts affording new forms of interaction and collaboration between humans and technology (Baird & Maruping, 2021). AI is endowed with the capacity to act autonomously, make decisions, and learn from their interactions. For individual practitioners, this means that they can delegate complex decision-making tasks to AI. However, studies have urged caution regarding accountable and responsible decision-making with AI, emphasising the need for careful consideration in maintaining control over delegation and ensuring the interpretability of decisions (Schwartz & Te'eni, 2024).

Now, to further discuss AI's role, it becomes imperative to understand the decision-making tasks in professional services. Decision-making tasks in professional services include diagnosis, inference, and treatment (Abbott, 1988), among which inference is the crux of the process, while diagnosis and treatment act as the intermediary conduits, transmitting information in and out of the professional

knowledge system (Spring et al., 2022). The uncertainties, uniqueness, and complexities of the professional service context require practitioners to apply their knowledge and make decisions adaptively to respond to evolving circumstances (Schön, 2017). Drawing upon these insights, our discussion now focuses on how to integrate AI into these decision-making tasks of professional service practitioners.

Diagnosis is the process of applying systemic knowledge to discern issues within a specific context. Previously, this was a human-led effort that leaned heavily on a professional's expertise to decipher and respond to contextual subtleties. The traditional IT philosophy, which views AI tools merely as mechanisms to enhance information processing and channel data into individual knowledge systems (Spring et al., 2022), fails to recognise the limitations of individual knowledge and thus constrains AI's full potential.

We advocate for an AI-proactive approach in the diagnostic phase, where AI is not just assimilating information but also proactively participating in defining and discovering problems. This involves employing AI 'knowledge' to identify anomalies and irregularities that may signal issues. Utilising natural language processing to find inconsistencies in documents for fraud detection (Craja et al., 2020) or using machine vision to detect illicit behaviour in auditing interviews (Pentland et al., 2017) are examples of this expansive role. In fact, studies demonstrate that AI can exceed practitioners' performance in discovering problems (Brinker et al., 2019), validating an AI-led approach to diagnosis. However, this approach has challenges to be addressed. AI, trained on universal data, may lack the capability to contextualise its 'knowledge' to specific cases (Raisch & Krakowski, 2021), leading to biased judgment and potentially overwhelming 'information bombardment' from too many false positives. This is where the practitioner's role should pivot to being contextual interpreters. They would provide AI with contextual insights explaining why certain patterns may appear anomalous but are in fact normal within the given context, and vice versa. Such a synergy of AI's analytical abilities with the practitioner's contextual understanding can lead to a more nuanced and effective professional diagnostic approach.

Inference in decision-making can be understood as the process of drawing conclusions based on evidence and reasoning. Existing literature has tended to position AI as a tool for automating simple inferences, while merely assisting with more complex inferences (Corea, 2019; Spring et al., 2022). However, this view overlooks the fundamental difference in rationality between AI and humans. AI operates on formal rationality, utilising data, algorithms, and mathematical models, whereas humans rely on substantive rationality, drawing on personal experience, intuition, and context (Lindebaum et al., 2020). This distinction becomes stark when contrasting inference models used by AI and humans in diagnosing discrepancies. AI, with its formal rationality, might detect an anomaly based on statistical deviation in a large dataset, whereas a human practitioner, using substantive rationality, might detect the same anomaly where the evidence does not match their initial impression.

These two forms of rationality ought to be viewed as complementary, rather than seeing the inference process as necessarily human-led with selective reliance on AI. The ideal approach would involve parallel processing of inferences by both AI and humans, followed by a comparative analysis of the conclusions reached by both parties. Discrepancies between the two conclusions could then be scrutinised, leading either to a correction of the human's inference or a recalibration of the AI's data inputs. This complementary approach in inference also addresses the black-box decision making mentioned earlier. Since it is very challenging, if not impossible, to fully understand AI's inference due to its algorithmic complexity, this complementary approach allows for a critical and informed review of AI's output rather than process. Through engaging in a dual process of inference, humans can spot inconsistencies or errors that the AI may not identify due to its lack of contextual understanding. We argue for the importance of combining the strengths of both AI's formal rationality and human's substantive rationality, enhancing the accuracy and robustness of the inference phase.

Treatment is the implementation of a solution to rectify the problem identified during the diagnosis and inference stages. It is the actualisation of a planned response to mitigate the identified issue. In our current regulatory framework, it is practitioners who bear the ultimate responsibility for the treatment provided (Munoko et al., 2020). For instance, auditors, rather than AI tools, are legally liable for the auditing opinion they sign. Other professions may also encounter similar issues, such as medical practitioners are accountable for the medicines they prescribe. This 'ultimate responsibility' can sometimes lead to practitioners being less willing to accept AI-driven advice on treatment, especially when it contradicts their own judgement (Commerford et al., 2022; Dietvorst et al., 2015, 2018). This hesitation might be even more pronounced among professional practitioners who are more confident in their own abilities and expertise (Logg et al., 2019). However, discounting AI's advice on treatment often results in undesirable outcomes.

It is important to note that humans often lack 'meta-knowledge' regarding where their strengths lie and in which areas AI prevails (Fügener et al., 2022). Practitioners may overestimate their own expertise while underappreciating the value of AI advice. However, there is growing evidence that the outcomes are largely improved when human do incorporate AI advice into their decision-making processes (Fügener et al., 2022; Logg et al., 2019). Therefore, we argue that a re-evaluation of the human-centric treatment is needed. Practitioners must better acknowledge the capabilities and limitations of both human and AI in providing treatment. A cooperative rather than competitive relationship between humans and AI should be cultivated, where AI's suggestions are seen as valuable contributions to decision-making rather than threats to professional autonomy.

Practical Implications for Professional Service Firms: Task Automation and Augmentation with AI.

Our findings reinforce the literature's view of AI as a tool for enhancing process automation and standardisation, boosting workforce capability, diversifying talent pools, and proposing new business models. However, our analysis also reveals that AI may introduce performance fluctuation, responsibility gaps, and replicate existing practices. Considering these constraints, some studies advocate for a pragmatic approach: using AI to automate well-defined, repetitive tasks and to augment complex tasks where AI's consistency might become a concern. (Eulerich et al., 2022; Sampson, 2021). Yet, this viewpoint has faced scepticism. An emerging discourse posits that automation and augmentation are not isolated paradigms but intertwined processes (Huang & Rust, 2018; Raisch & Krakowski, 2021), and firms should take a balanced and dynamic view on AI automation and augmentation (Sabherwal & Grover, 2024). Raisch's paper (2021) cites examples of both JP Morgan and Symrise, where previously augmented processes have been eventually automated (p. 196–197). The authors argue that automation can liberate resources for augmentation, while augmentation

can help discover the rules or models that enable automation. Guided by this dynamic perspective, we challenge the static assumption of choosing between automation and augmentation based on task complexity and repetitiveness. Instead, we advocate for a dynamic, holistic strategy that incorporates inherent attributes of professional services such as knowledge intensity and high customisation (von Nordenflycht, 2010; von Nordenflycht et al., 2015). This approach facilitates a flexible transition between automation and augmentation, addressing the tension among automation, consistency, and standardisation.

The transition from **automation to augmentation** is a process where the automation of one task can influence and enhance the augmentation of adjacent tasks (Raisch & Krakowski, 2021). To optimise cost-efficiency, firms often initially target high-volume, repetitive, and low-judgemental tasks in which AI usually delivers consistent outcomes (Eulerich et al., 2022). For instance, in the legal profession, AI can automate case research or contract review, while in healthcare, AI can automate the extraction of information from medical report and patient's history. The automation of such processes significantly standardises and structures the raw information. For instance, automating the collection of patient health information in a clinical setting can result in a uniform, structured database of patient details for further use. Once automation is achieved, the structured data can be fed into more strong AI systems such as deep learning for further analysis. The output of this analysis serves to enhance, not replace, practitioners' judgement in areas that require high levels of professional knowledge. For example, in the legal profession, the deep learning AI might provide predictions about case outcomes based on historical data (Sukanya & Priyadarshini, 2021), thus aiding lawyers in their strategy formulation. This process of using AI to bolster professional judgement is referred to as AI-augmentation of tasks. This shift from automation to augmentation, although seemingly straightforward, necessitates strategic planning. With basic tasks automated, the focus can then transition to enhancing adjacent, more intricate professional tasks through AI.

The transition from **augmentation to automation** involves the automation of tasks that were previously AI-augmented, typically requiring a high degree of professional knowledge and judgement. This move is not without challenges, particularly due to the high level of customisation inherent in professional services. Indeed, as our findings highlight, such variability makes standardisation difficult, evidenced by the fluctuating performance of strong AI across different scenarios (Craja et al., 2020; Dong et al., 2018). However, this complexity does not necessarily preclude the possibility of transition. Over time, the close collaboration between practitioners and AI may expose patterns or models within the variability of individual cases, shedding light on commonalities and distinctions, and subsequently guiding the development of more adaptable, intelligent automation solutions. JP Morgan and Symrise, as Raisch and Krakowski (2021) noted, made the case.

In essence, this transition hinges on a symbiotic relationship between AI and practitioners within the firm. As AI distils patterns from data, practitioners apply and test these patterns in specific cases, providing AI with feedback. Where patterns fall short, practitioners' context-rich feedback aids AI in pattern refinement. This iterative process opens the possibility of automating high-judgment tasks with high consistently outcomes across varying scenarios. This process, while requiring a strategic approach and time, ultimately holds significant potential to revolutionise the landscape of professional services firms.

Practical Implications for the Professional Association: Recalibrate the Knowledge System with AI.

Our findings indicate that AI affords professional associations the chance to claim their legitimacy and update their knowledge systems, yet it may also impose constraints as de-professionalisation. Faced with the dual task of securing legitimacy through enhanced service quality while countering legitimacy erosion from de-professionalisation, professional associations must harness the power of AI in refining what underpins the legitimacy: the professional knowledge system. The crux of professional services rests heavily on specialised knowledge and expertise (Abbott, 1988), an asset that bestows the profession with a distinctive legitimacy and authority and enables it to competently perform specialised tasks (Schön, 2017). However, the ascent of AI presents challenges to the role of professionals in knowledge creation and application, where AI can undertake knowledge tasks traditionally demanding human judgment and creativity (Benbya et al., 2024). This development raises questions about how to integrate human-derived tacit knowledge with AI-generated explicit knowledge effectively (Alavi et al., 2024). Professional associations – recognised for their pivotal role in capturing, codifying, and normalising knowledge into practice (Greenwood et al., 2002), and further disseminating new knowledge across the profession (Amin & Roberts, 2008) – are strategically positioned to recalibrate their knowledge systems with AI. We therefore propose that professional associations should embrace AI as a catalyst for the creation, refinement, and retirement of knowledge.

Knowledge creation with AI starts with leveraging advanced AI tools to process vast amounts of raw data, identifying patterns that may not be readily apparent or previously known. These detected patterns, however, do not constitute knowledge by themselves. The transformation of these patterns into actionable knowledge is where the role of professional associations becomes crucial. Professional associations function as pivotal platforms for practitioners to convene, share experiences, and cultivate new understandings of effective practice (Greenwood et al., 2002). As AI unveils novel patterns, these associations could orchestrate practitioner communities to scrutinise and authenticate these patterns across various contexts. Through such collaborative endeavour, the tacit potential of AI-generated patterns is validated and interpreted for application (Amin & Roberts, 2008). Subsequently, professional associations codify the validated insights into industry standards and guidelines for actions. With new knowledge created by AI, professional associations not only ensure the profession's continued relevance, but also diversify professional roles emerged from new knowledge, countering the task displacement impact.

Knowledge refinement with AI embarks on a journey of challenging existing knowledge systems, interrogating taken-for-granted assumptions, and unveiling new truth. While literature suggests that professional associations once acted as impediments to radical knowledge transformations, favouring only incremental shifts (Amin & Roberts, 2008), it should no longer be the case in the AI era. Research like Goto's (2022) highlights that professional associations are fostering a shared vision among professionals, acknowledging that AI-driven advancements warrant substantial alterations to the profession's nature. This collective understanding forms a fertile ground for professional associations to propose revisions to the existing knowledge system. The refinement process commences with

Table A1
Distribution of Articles Based on Focal Technology.

<i>Focal technology</i>	<i>Count</i>
AI (MLNA)	26
CAATTs (GAS)	15
Unspecified AI	12
AI (ES)	6
AI (RPA)	6
CAATTs (CAS)	6
AI (NLP)	5
AI (MLNA and NLP)	5
Unspecified CAATTs	5
AI (ES and MLNA)	3
AI (CV)	2
AI (CV and NLP)	1
Grand total	92

professional associations collecting AI-discovered patterns from the field. Then professional associations can synthesis the patterns to generate higher levels of insight to scrutinise the existing knowledge corpus, pinpointing potential misconceptions and proposing changes. AI-refined knowledge enhances service quality and relevance, thereby facilitating the association's claim for legitimacy.

Knowledge retirement with AI signifies another critical knowledge recalibration process for professional associations addressing task displacement by automation. It involves discerning knowledge and skills that, while accurate, have become less relevant due to tasks being automated. For instance, with AI taking over basic accounting tasks, understanding the manual process of ledger entry may lose relevance for accountants in an AI-driven environment. In this context, professional associations need to have an in-depth understanding of processes that can function autonomously, without human intervention at the field level (Huang & Rust, 2018). Subsequently, they should identify the pertinent knowledge and skills related to these processes and initiate their retirement. This may involve eradicating such knowledge from professional guidelines and educational prerequisites. Undertaking this process enables professional associations to bolster legitimacy by ensuring that the profession remains agile and responsive to the evolving landscape of the industry.

To summarise, our findings underscore the urgency and importance of embracing AI within professional services, necessitating strategic reforms at the levels of individual practitioners, firms, and professional associations. Effective harnessing of AI's potential and successful navigation of its challenges would require a refined approach to decision-making, a dynamic view on automation and augmentation, and a recalibration of the knowledge system. This transformative journey, although complex, is essential for individuals, firms, and professional associations in the contemporary professional landscape.

Conclusion

This developmental literature review paper uses TACT as a theoretical lens to delve into the use of AI in a professional service industry. Based on 92 papers in high-ranked journals, we conceptualised the affordances and constraints of AI tools, comparative to classic IT, in terms of enabling and restricting actions for individuals, firms, and professional associations within professional services. Such conceptualisation formed the basis for articulating the strategic implication for those entities.

This study makes several theoretical contributions to the IS discipline. The first contribution is synthesising existing literature to produce knowledge of AI from a socio-technical perspective. The conceptualisation of AI affordances and constraints provides insights into the potential positive and negative consequences of using AI by social actors at multiple levels. Our proposed affordance and constraints framework complements the current discussion on AI in relation to its interaction with various social groups (Dwivedi et al., 2021). Moreover, our discussion reveals the reciprocal and tensional relationship between affordances and constraints across different levels of actors. It expands the current notion of deviant affordance (Haag et al., 2022), inviting a more profound exploration of the ways in which IS technologies are used and managed in conflicting sociotechnical relationships (Asatiani et al., 2021). Lastly, this study extends the applicability of TACT to a broader context. Prior literature has primarily employed TACT to study the consequences of the use of IT artefacts by individuals and organisations (Du et al., 2019; Leidner et al., 2018; Leonardi, 2011). This study is among the first to apply TACT to investigate the distinctions between two types of artefacts, as a foundation for articulating the strategic implications of technological change. The study may provide useful insights for future research aiming to use TACT to investigate similar IS phenomena.

While our research is contextualised within auditing, the findings and implications are believed to be transferable to a wider array of professional service industries, such as legal and medical practices, due to their significant commonalities in terms of organisational structures, client interactions, knowledge intensity, and customisation (Spring et al., 2022). Our study suggests a symbiotic integration of AI within the decision-making, task execution, and knowledge systems of the professional services landscape, calling for paradigm shift at the levels of individual practitioners, firms, and professional associations. Individual practitioners are urged to reconsider their roles in decision making, adopting a collaborative stance with AI in diagnosis, inference, and treatment. This convergence of human judgement and AI analysis optimises accuracy and enhances the quality of decision-making. Concurrently, professional service firms should adopt a dynamic approach to AI strategy, fluidly transitioning between task automation and augmentation. This allows for continual reconfiguration based on the evolving needs and circumstances within the professional service firms. Lastly, professional

Table A2
Summary of Studies in the Literature Review.

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
Abdolmohammadi (1991)	Mixed CAATTs	Exploratory: explore auditors' perceptions of CAATTs use and importance.	Task structure theory	Quantitative analysis of survey data	Auditors value CAATTs highly, especially in Big 4 firms where its use and importance are perceived more significantly, correlating positively with the auditors' IT expertise, but not for auditors in small firms with limited IT usage.
Abou-El-Sood et al. (2015)	CAATTs (GAS)	Explanatory: evaluate the nature and extent of GAS used by auditors and examine factors influencing its use.	IT Audit Process Quality Framework	Quantitative analysis of survey data	A significant 73 % of UK auditors do not use GAS, mainly due to its perceived limited benefits for auditing small clients, high implementation costs, a substantial learning curve, and preference for traditional manual methods.
Ahmi and Kent (2012)	Mixed AI	Theory building: introduce affect-based trust and work system model into the AI-supported auditing process.	Affect-based trust; affective work system	Conceptual analysis	As AI advances, it will play a more significant role in the audit process, potentially leading to a shift in the way audits are conducted and perceived in terms of trust and credibility.
Alles and Gray (2019)	Mixed AI	Explanatory: examine factors influencing auditors' choice for AI tools for various audit tasks.	Task structure theory	Quantitative analysis of survey data	The choice of AI tools by auditors for verifying financial information is influenced by the task's routineness, their job level, and skills, with auditors in semi-routine firms and higher job positions preferring these tools more.
Alles et al. (2004)	CAATTs (CAS)	Design science: propose tertiary monitoring solutions to enhance audit effectiveness and transparency.	Continuous audit concept	Design science method	It suggests that implementing a "black box" log file system, which records auditors' actions and decisions, could significantly enhance the credibility of audits by providing a detailed and tamper-proof audit trail
Alles et al. (2006)	CAATTs (CAS)	Design science: implement a proposed CAS for business process controls at Siemens.	Continuous audit concept	Design science method	CAS has potential for cost savings and improved audit quality through automation of audit processes, though challenges such as the formalisation of audit procedures and the management of audit alarms are highlighted.
Back et al. (2001)	AI (MLNA and NLP)	Explanatory: compare, cluster, and analyse quantitative and qualitative data from annual reports using a subset of neural networks.	Machine learning theory	Quantitative and qualitative analysis of archival data	Clusters generated from textual information do not always align with those based on quantitative data, suggesting that qualitative and quantitative data offer complementary insights for auditors into a company's performance and outlook.
Baldwin-Morgan (1995)	AI (ES and MLNA)	Normative: expert systems and neural networks should be included in accounting education because of their growing practical importance.	N.a	Conceptual analysis and prescriptive	Neural networks and ES have potential to simulate human decision-making and pattern recognition, enhancing tasks like auditing and financial prediction.
Bertomeu et al. (2021)	AI (MLNA)	Explanatory: evaluate the performance of machine learning in detecting and interpreting patterns in accounting misstatements.	Machine learning theory	Quantitative analysis of archival data	MLNA, particularly GBRT, outperforms traditional logistic regression models in detecting misstatements, with notable effectiveness in predicting future misstatements and identifying severe cases.

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Table A2 (continued)

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
Bhattacharya et al. (2011)	AI (MLNA)	Design science: explore the application of Benford's law in financial fraud detection and designs a neural network-based decision support system.	Machine learning theory	Quantitative analysis of archival data	ANN can reduce critical classification errors that lead to audit failures, demonstrating the potential of integrating ANNs and Benford's law into auditor decision support systems for improved fraud detection.
Bierstaker et al. (2014)	CAATTs (GAS)	Explanatory: examine factors influencing auditors' use of CAATTs.	Unified theory of acceptance and use of technology	Quantitative survey analysis	Auditors' use of CAATTs is influenced by performance expectancy, organisational pressures, and the technical infrastructure's support
Boritz and Stratopoulos (2022)	AI (RPA)	Exploratory: identify future research avenues within RPA in accounting and auditing.	N.a	Qualitative analysis of workshop data	RPA can automate routine tasks, allowing auditors to engage in more cognitive work; but it needs to have sound governance to prevent "RPA hell," and ensure the auditability of automated systems.
Bradford et al. (2020)	CAATTs (GAS)	Explanatory: examine factors influencing auditors' perceptions of GAS use and audit benefits.	DeLone and McLean's information systems success model	Quantitative survey analysis	For financial auditors, detecting material misstatements is the primary indicator of GAS usefulness, whereas IT auditors value GAS for its capabilities in detecting control deficiencies and fraud, impacting their perception of information quality.
Braun and Davis (2003)	Mixed CAATTs	Descriptive: describe auditors' perceptions on advantages and disadvantages of using CAATTs in audit process.	N.a	Quantitative survey analysis	While auditors recognise the benefits of GAS for improving audit outcomes, they express a need for more training to enhance their technical proficiency and confidence in using these tools.
Brown et al. (2020)	AI (NLP)	Explanatory: evaluate thematic content using LDA predicts intentional financial misreporting.	Machine learning theory	Qualitative and quantitative analysis of archival data	Thematic content, when added to models based on financial and textual variables, significantly improves the detection of financial misreporting, highlighting the value of analysing the thematic aspects of disclosures beyond conventional metrics.
Busta and Weinberg (1998)	AI (MLNA)	Explanatory: use Benford's distribution and artificial neural networks to detect normal vs. manipulated financial data.	Machine learning theory	Quantitative analysis of simulated data	Neural network can correctly identify manipulated data sets with a significant level of accuracy, suggesting this method's potential as a tool for auditing and fraud detection.
Calderon and Cheh (2002)	AI (MLNA)	Exploratory: potential applications of neural networks in the fields of auditing and risk assessment.	Machine learning theory	Literature review, conceptual analysis	Neural network can enhance business risk-based auditing by accurately assessing various risk categories, such as control risk, information risk, and fraud detection.
Caster et al. (2021)	AI (RPA)	Exploratory: explore the changes in bank confirmation processes over time as introducing RPA.	N.a	Qualitative analysis of case study data	A significant increase in the use of electronic confirmations, with errors requiring reconfirmation being less than 2 percent of all electronic confirmations.
Charitou et al. (2004)	AI (MLNA)	Explanatory: use logit and neural networks to predict corporate failures in UK.	Machine learning theory	Quantitative analysis of archival data	An NN-logit based model including operating cash flow, profitability, and financial leverage variables achieved an overall correct classification accuracy of 83 % one year prior to failure.

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Table A2 (continued)

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
Chen et al. (2017)	AI (NLP, MLNA)	Design science: design and develop a solution for detecting fraud using NLP and other ML algorithms.	Machine learning theory	Design science method	The proposed fraud detection method combines text based and numerical analysis, enhancing the accuracy of identifying fraudulent narratives in annual reports.
Christ et al. (2021)	AI (CV)	Design Science: design and develop drones and CV to enhance audit quality in inventory count.	Machine learning theory	Design science method	Using drones and CV improves the efficiency and accuracy of inventory audits compared to manual methods. The technology also provides superior audit documentation, but its adoption faces institutional barriers.
Chung-Fern Wu (1994)	AI (MLNA)	Explanatory: apply artificial neural networks to capture and utilise senior auditing expertise effectively.	Machine learning theory	Quantitative analysis of archival data	ANNs can effectively learn and replicate the decision-making patterns of experienced auditors, offering a promising approach to preserving and utilising expert knowledge in auditing practices, especially in scenarios where direct knowledge transfer is challenging.
Commerford et al. (2022)	AI (MLNA)	Explanatory: investigate algorithm aversion in auditor judgments in a complex estimate task.	Algorithm aversion	Quantitative analysis of experiment data	Auditors are likely to propose smaller adjustments to management's estimates when contradictory evidence is provided by an AI system compared to a human specialist. This effect is amplified when management uses more objective inputs for their estimates.
Cooper et al. (2019)	AI (RPA)	Descriptive: showcase the application of RPA within a case firm.	N.a	Qualitative analysis of case study data	The implementation of RPA has led to increases in efficiency and quality. Despite efficiency gains, firms are not reducing headcount but are instead reallocating employees to higher-value tasks.
Craja et al. (2020)	AI (MLNA and NLP)	Explanatory: evaluate the performance of deep learning for detecting financial statement fraud combining financial and textual data.	Machine learning theory	Qualitative and quantitative analysis of archival data	The combined analysis of financial and textual data using deep learning significantly improves the detection of fraudulent financial statements, with the model providing interpretable indicators that can aid stakeholders in fraud detection.
Curtis and Payne (2008)	Mixed CAATTs	Explanatory: examine contextual factors and individual characteristics influencing auditors' decisions to implement CAATTs in auditing.	Unified theory of acceptance and use of technology	Quantitative analysis of experiment data	Firms can influence CAATTs adoption decisions by adjusting budget and evaluation periods and by communicating approval from higher management. Individual characteristics of auditors, like risk-aversion and perceptions of budgetary pressure, also determine their adoption decisions.
Curtis and Payne (2014)	Mixed CAATTs	Explanatory: examine technology acceptance in audit settings under budgetary pressure and different social influences.	Unified theory of acceptance and use of technology	Quantitative analysis of survey data	Higher levels of performance expectancy, influenced by effort expectancy and facilitating conditions, are associated with increased intentions to use CAATs, but the significant varies under different budgetary pressure.
Davis et al. (1997)	AI (ES and MLNA)	Design science: design, develop and evaluate an integrated AI system (ES and ANN) for	Machine learning theory; decision theory	Design science method	Integrating expert systems with neural networks can significantly support auditors in complex judgment tasks, such as assessing

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Table A2 (continued)

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
		supporting complex audit judgment tasks.			control risk, by leveraging both explicit audit knowledge and implicit pattern recognition capabilities
Debreceny and Gray (2011)	AI (NLP)	Design science: design and evaluate an audit process using NLP to indicate fraud by checking emails.	Machine learning theory and text mining concept	Design science method	Using NLP to detect potential fraud in email communication could potentially increase auditing scope and enhance quality.
Debreceny et al. (2005)	CAATTs (GAS)	Exploratory: explore the application and implications of using GAS in the financial services sector, particularly banks.	N.a	Qualitative analysis of interview data	External auditors reported not using GAS, attributing this to the tools' perceived inapplicability to the types of tests required for financial statement assertions and internal controls at banks.
Dillard and Yuthas (2001)	AI (ES)	Normative: establish a set of ethics for the implementation and use of audit expert systems.	Niebuhr's theory of responsibility	Conceptual analysis and prescriptive	Advocating for a responsibility ethic in audit expert systems, emphasising the importance of considering the ongoing relationships and effects of these systems on all stakeholders involved.
Ding et al. (2020)	AI (MLNA)	Explanatory: evaluate the effectiveness of machine learning models in improving the accuracy of accounting estimates specifically for insurance loss reserves	Machine learning theory	Quantitative analysis of archival data	Machine learning models, on average, produce more accurate loss estimates than those made by managers, suggesting the potential for these technologies to significantly enhance the reliability and usefulness of financial information.
Dong et al. (2018)	AI (NLP and MLNA)	Explanatory: evaluate the effectiveness of NLP in leveraging social media data for corporate fraud detection	Systemic functional linguistics theory	Qualitative and quantitative analysis of archival data	NLP leveraging social media data provides valuable signals for detecting corporate fraud, which outperforms baseline methods using only financial ratios or textual features from traditional financial documents.
Elam and Mead (1990)	CAATTs (GAS)	Explanatory: evaluate the effectiveness of a decision aid in enhancing auditors' creativity of decision making.	Amabile's componential model of creativity	Quantitative analysis of experimental data	The use of creativity-enhancing decision aids can significantly affect the creativity of auditors. However, decision aids design can both enhance and inhibit creativity, depending on how it is perceived and used by auditors.
Elliott (1986)	AI (ES)	Descriptive: describe and discuss the reasons behind the expansion of auditing scope and the adoption of new technologies like ES.	Attestation concept	Describe and conceptual analysis	Auditors need to broaden their scope to new information types, incorporate ES for quality and productivity, and become proactive "information people" to navigate the technological shifts.
Etheridge et al. (2000)	AI (MLNA)	Explanatory: evaluate the performance of different artificial neural network models in assisting auditors to evaluate client financial viability.	Machine learning theory	Quantitative analysis of archival data	Probabilistic neural network model was most reliable in classification based on overall error rates, while categorical learning network model was least costly when considering misclassification costs.
Eulerich et al. (2022)	AI (RPA)	Design science: develop and validate a three-step framework for auditors in deciding which activities to automate using RPA.	Socio-technical systems theory	Design science method	To implement RPA, auditors need to identify repetitive, rule-based tasks suitable for automation, and assess the technical feasibility and potential return on investment of automating selected tasks.
Farkas and Hirsch (2016)	CAATTs (GAS)	Explanatory: examine the impact of the frequency and automation of internal control	Strength of internal audit function model	Quantitative analysis of experimental data	Automated remediation strategies fully remediate auditors' perceptions of poor IAF

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Table A2 (continued)

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
		testing (by GAS) on auditors' reliance on the internal audit function (IAF).			work performance and low degree of reliance on the IAF, whereas manual remediation strategies result in only partial remediation.
Fedyk et al. (2022)	Mixed AI	Explanatory: first to explore then to examine the impact of adopting AI on firm performance and employment changes.	Babina's AI-human capital framework	Qualitative analysis of interview data and quantitative analysis of archival data	Investing in AI helps firms improve audit quality, reduces fees, and ultimately, displaces human auditors, although the effect on labour takes several years to materialise.
Fischer (1996)	CAATTs (GAS)	Theory building: theorise the way in which CAATTs bring benefit to auditors and auditing firms.	Theory of social constructivism	Qualitative analysis of field and interview data	The benefits of new audit technologies, in terms of audit efficiency, do not arise directly from the technologies themselves but rather through a reduction or elimination of other audit procedures.
Goto (2021)	Mixed AI	Theory building: theorise AI's impacts on collective professional role identity among auditors.	Role identity theory	Qualitative analysis of archival and interview data	AI's adoption is leading to the construction of a new collective professional role identity among auditors, through enhancing managerialism and professionalism, and sustaining legal legitimacy.
Goto (2022)	Mixed AI	Theory building: theorise auditors' sense-making about AI and its impact on their future work.	Sensemaking theory	Qualitative analysis of archival and interview data	Auditors' sensemaking about AI involved monitoring changes, experimenting with practices, and future blueprints.
Green and Choi (1997)	AI (MLNA)	Explanatory: evaluate neural network models' effectiveness in classifying financial statements for fraud detection.	Machine learning theory	Quantitative analysis of archival data	Neural networks, particularly those developed with the plain sum-of-years'-digit weighted average method, significantly reduce Type I and II errors in fraud detection.
Hajek and Henriques (2017)	AI (MLNA, NLP)	Explanatory: compare machine learning methods for detecting financial statement fraud using financial and linguistic data from corporate annual reports.	Machine learning theory	Quantitative and qualitative analysis of archival data	Ensemble methods excel in identifying fraudulent firms with high true positive rates, while Bayesian belief networks perform best in identifying non-fraudulent firms.
Hansen and Messier Jr (1986)	AI (ES)	Design science: create and demonstrate a knowledge-based expert system for auditing computer systems.	Decision theory	Design science method	A prototype expert system, based on existing technology, can effectively support computer audit specialists in evaluating the reliability of a client's computer system through a network of decision rules.
Hansen et al. (1992)	AI (MNLA)	Explanatory: evaluate machine learning algorithms against generalised qualitative-response models for audit decision-making domains.	Machine learning theory	Quantitative analysis of archival data	Qualitative response models outperformed ID3 and ANN models; EGB2 model showed the highest classification accuracy in audit opinion.
Hardy and Laslett (2015)	CAATTs (CAS)	Exploratory: investigate the practicalities and visibility of implementing CAS in auditing.	Continuous audit concept	Qualitative analysis of case study data	Case firm implemented over 100 automated tests daily with predictive analytics, integrating exception management systems and visualisation technology.
Henderson et al. (2016)	CAATTs (GAS)	Explanatory: examine the enablers and inhibitors of GAS and their effect on GAS usage.	Dual factor theory	Quantitative analysis of survey data	Perceived threat and system problems significantly inhibit GAS usage, with the effects of system problems fully mediated by perceived usefulness and perceived ease of use, and the effect of perceived threat being partially mediated by perceived usefulness.

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Table A2 (continued)

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
Hian and Chan (2004)	AI (MLNA)	Explanatory: compare neural networks, decision trees, and logistic regression for going concern prediction.	Machine learning theory	Quantitative analysis of archival data	Decision tree models outperform logistic regression and neural network models in predicting going concern status, showcasing its effectiveness in analysing complex, non-linear relationships in financial data
Huang and Vasarhelyi (2019)	AI (RPA)	Design science: develop and evaluate a framework for applying RPA in auditing practices	Audit procedure automation framework	Design science method	The successful application of RPA can significantly enhance audit efficiency and effectiveness by automating routine tasks, thus allowing auditors to focus on areas that require expert judgment.
Huang et al. (2009)	CAATs (GAS)	Design science: develop and validate a mechanism for detecting gaps between information system process flows and internal control flows.	Dynamic model	Design science method	The developed mechanism can effectively identify discrepancies between the designed internal control flows and the actual processes within information systems, thereby aiding in enhancing data quality and ensuring compliance with internal controls.
Humpherys et al. (2011)	AI (NLP)	Explanatory: evaluate linguistic credibility analysis techniques to detect fraudulent financial statements.	Information manipulation theory	Qualitative analysis of archival data	Fraudulent disclosures use distinct linguistic patterns compared to non-fraudulent ones, including more activation language and less lexical diversity.
Koh and Tan (1999)	AI (MLNA)	Explanatory: evaluate the effectiveness of neural networks in predicting firms' going concern status.	Machine learning theory	Quantitative analysis of archival data	Neural network model accurately predicted the going concern status of all test cases in a hold-out sample, suggesting that neural networks could offer a promising alternative to traditional statistical models for going concern prediction.
Koskivaara (2000)	AI (MLNA)	Explanatory: evaluate the effectiveness of artificial neural network models in auditing monthly balances.	Machine learning theory	Quantitative analysis of archival data	Neural networks are promising tools for recognising the dynamics and relationships between financial accounts, offering potential improvements in audit efficiency and accuracy by identifying unusual fluctuations and predicting future account values.
Koskivaara (2004)	AI (MLNA)	Exploratory: explore the application of artificial neural networks in auditing, specifically in analytical review procedures	Machine learning theory	Literature review, conceptual analysis	ANNs have potential to significantly support and enhance analytical review procedures in auditing by learning from examples, recognizing complex patterns, and comparing them with actual values, offering a promising tool for auditors to handle increasingly complex audit tasks
Krambia-Kapardis et al. (2010)	AI (MLNA)	Explanatory: test the effectiveness of artificial neural networks for fraud detection.	Machine learning theory	Quantitative analysis of survey data	ANNs can predict fraud with 90 % accuracy, demonstrating their potential utility in helping auditors identify companies at risk of fraud at planning stage.
Krieger et al. (2021)	Mixed AI	Theory building: theorise the process of AI adoption in auditing firms, factors influencing the process, and adoption outcomes.	Process theory	Qualitative analysis of interview data	AI adoption process involves ideation, evaluation, commitment, solution building, deployment, and use, where technological capabilities of audit firms are critical for this process.

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Table A2 (continued)

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
Lehner et al. (2022)	Mixed AI	Normative: identify ethical challenges of using AI-based systems for decision-making, suggesting implications for auditing scholars and practitioners.	Rest's model of antecedents for ethical decision making	Literature review	Five major ethical challenges of AI-based decision-making: objectivity, privacy, transparency, accountability, and trustworthiness. AI alone cannot make ethical decisions due to the lack of necessary preconditions.
Lenard et al. (1995)	AI (MLNA)	Explanatory: evaluate the effectiveness of neural networks and logit models to improve the auditor's going concern judgement.	Machine learning theory	Quantitative analysis of archival data	GRG2-based neural network predicts going concern uncertainty with 95 % accuracy; the logit model's prediction accuracy was also high, but it had difficulties converging with a smaller set of variables.
Lenard et al. (2000)	AI (MLNA, ES)	Explanatory: evaluate the use of fuzzy clustering and a hybrid model combining statistical models and expert systems for auditors' going concern assessments	Machine learning theory	Quantitative analysis of archival data	Both fuzzy clustering and the hybrid model effectively differentiate between firms likely to face going concern issues and those that are not, but the latter comes with greater accuracy.
Li et al. (2007)	CAATTs (CAS)	Design science: propose and validate a methodology for developing a CAS based on information process models	Continuous audit concept	Design science method	CAS effectively assists auditors in performing tasks by enabling them to design auditing rules and embed these rules into the database system. This approach facilitates real-time and continuous monitoring of transactions, thereby enhancing the auditors' ability to detect anomalies and improve audit efficiency.
S. Li et al. (2021)	AI (MNLA)	Explanatory: examine the effectiveness of artificial neural networks in analytical procedures for auditing	Machine learning theory	Quantitative analysis of simulation data	Artificial neural networks, while not significantly outperforming traditional methods in overall effectiveness, show a considerable reduction in Type II errors.
Liang et al. (2001)	CAATTs (GAS)	Design science: create a new CAATTs system that leverages emerging technologies, like intelligent agents, for auditing data processing systems.	Electronic auditing framework	Design science method	The proposed system provides a more effective, automatic, and electronic means of auditing, highlighting its advantages such as interoperability across systems, direct access to live data, and enhanced security and fraud detection capabilities.
Lin and Wang (2011)	Mixed CAATTs	Theory building: construct an auditing software assessment model.	Analytic network process	Qualitative analysis of interview data	System functions, data processing, technical support, and cost are critical criteria for auditing software selection, with system stability, data accuracy, and technical support being particularly significant factors.
Lin et al. (2015)	AI (MLNA)	Explanatory: examine the effectiveness of various ML techniques compared to expert judgments in detecting financial statement fraud.	Fraud triangle theory	Quantitative analysis of archival data	ANNs and CART have higher correct classification rates compared to logistic regression. Additionally, the type II error for ANNs significantly drops, indicating a lower likelihood of failing to detect fraud when it is present.
Liu et al. (2022)	AI (NLP)	Explanatory: examine the associations between Key Audit Matters (KAM) sentiment in audit reports and firm performances.	Accountability theory; machine learning theory; Signalling theory	Quantitative and qualitative analysis of archival data	KAM sentiment is positively associated with future firm financial and market performance; The use of BERT for automatic sentiment extraction is validated as a

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Table A2 (continued)

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
Manita et al. (2020)	Mixed AI	Exploratory: explore the impact of digital transformation on external audit and its role in corporate governance.	Agency theory and stakeholder theory	Qualitative analysis of interview data	method for textual mining in audit reports. AI Digitalization improves audit relevance, quality, and governance mechanisms; for individual auditors, AI enhances decision-making, risk evaluation, and fraud detection.
Manson et al. (2001)	CAATTs (GAS)	Exploratory: explore audit automation brought by GAS as control, deskilling, resistance, and competition mechanisms within audit firms.	Giddens' structuration theory	Qualitative analysis of interview data	Audit automation enhances control, competitiveness, and managerial surveillance in audit firms, leading to opportunities for more skilled work for staff.
Munoko et al. (2020)	Mixed AI	Normative: highlight ethical implications related to the use of AI in auditing.	Anticipatory technology ethics framework	Conceptual analysis; bibliometric analysis	AI technologies, while offering significant benefits for auditing, also introduce ethical challenges related to safety, transparency, privacy, accountability, and user autonomy.
Nasir et al. (2021)	AI (MLNA)	Design science: develop a predictive framework and a system incorporating MLs to identify firms' internal control weaknesses.	Machine learning theory	Design science method	The proposed tool outperforms traditional models regarding internal control weakness detection in sensitivity and specificity after applying feature selection and data balancing techniques.
Nelson et al. (2000)	AI (ES)	Design science: develop and apply the EDGAR Agent, an expert system in the context of online continuous auditing.	Virtual auditing quality/ service framework	Design science method	The proposed system is capable of retrieving financial data from the database; challenges identified include handling the variety of accounting terms and financial statement formats and ensuring the speed and accuracy of data retrieval and analysis.
Pentland et al. (2017)	AI (CV)	Design science: develop a video-based screening system for risk assessment using nuanced facial features to detect deceit.	Machine learning theory	Design science method	Facial rigidity is a consistent indicator of deceit in controlled environments, offering a new approach for automated risk assessment in various fields, such as auditing and security screening.
Perols (2011)	AI (MLNA)	Explanatory: evaluate the effectiveness of statistical and machine learning algorithms for detecting financial statement fraud.	Machine learning theory	Quantitative analysis of archival data	Logistic regression and support vector machines perform well relative to more complex models like artificial neural networks in detecting financial statement fraud.
Santosoosso (2022)	AI (MLNA)	Explanatory: examine the effectiveness of ML algorithms leveraging micro in forecasting accuracy and analytical procedures.	Machine learning theory	Quantitative analysis of archival data	Neural network models provided more accurate forecasts than multiple regression models for both financial ratios tested, but also present overfitting issues.
Sarkar and Sriram (2001)	AI (MLNA)	Explanatory: examine the effectiveness of Bayesian models for early warning of bank failures.	Machine learning theory	Quantitative analysis of archival data	Using financial indicators, Bayesian models accurately predict bank failures using historical data.
Seethamraju and Hecimovic (2022)	Mixed AI	Exploratory: investigate the factors influencing the adoption of AI in auditing	Technology, organisation and environment framework	Qualitative analysis of interview data	Perceived benefits, compatibility, maturity, firm readiness, data quality, trust in AI technology, audit standards, regulation readiness, and clients' readiness play a significant role in the adoption process.
Shih (2010)	CAATTs (GAS)	Explanatory: examine the risk indicators for of financial examination.	Decision making trial and evaluation laboratory and analytic network process	Quantitative analysis of survey data	Together with five other factors, lack of a computer-assisted auditing system is one of the most important risk factors in financial examination.

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Table A2 (continued)

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
Siew et al. (2020)	CAATTs (GAS)	Explanatory: examine the factors influencing CAATTs adoption in developing countries like Malaysia.	Technology, organisation, and environment framework	Quantitative analysis of survey data	Factors influencing CAATTs adoption: client information system complexity, authority support, firm size, top management commitment, employee IT competency.
Singh and Best (2015)	CAATTs (CAS)	Design science: develop and implement a CAS in an SAP ERP environment to enhance audit efficiency and fraud detection.	Continuous audit concept	Design science method	CAS can significantly improve the detection of potential frauds and policy violations through continuous monitoring of transactions.
Singh et al. (2013)	CAATTs (CAS)	Exploratory: investigate the development and implementation of CAS in real-world settings.	Continuous audit concept	Qualitative analysis of case study data	Implementation of CAS provides continuous and near real-time surveillance of transaction data.
Sun (2019)	AI (NLP, CV)	Theory building: propose a framework for applying deep learning techniques to audit procedures	Audit process framework	Conceptual analysis	Deep learning can significantly enhance audit procedures by providing better information identification and judgment support across different audit phases.
Sutton et al. (2016)	Mixed AI	Descriptive: revisit the status of AI in accounting and auditing research, against previous assertions of its decline.	N.a	Literature review	A continuous increase in AI research within accounting over the past three decades shows the sustained interest and application of AI in accounting practices, contrary to earlier perceptions of a decline.
Tiron-Tudor and Deliu (2022)	Mixed AI	Exploratory: explore the evolving dynamics of human-algorithm interaction within the auditing process	Human in the loop system	Literature review and thematic analysis	Auditor-Governing-the-Loop is a potential future scenario, emphasising the strategic integration of human insight and algorithmic efficiency for enhancing audit practices.
Wang and Cuthbertson (2015)	AI (MLNA)	Commentary: highlight and propose areas within audit data analytics and AI that require further investigation.	N.a	Literature review and qualitative analysis of expert insight	Eight critical issues, such as factors affecting the interpretation of AI results and consequences of using AI, urgently require further research.
West and Bhattacharya (2016)	AI (MLNA)	Descriptive: describe the current domain of financial fraud detection using AI, covering various AI types, methods, and challenges of fraud detection	Machine learning theory	Literature review	AI-based techniques are highlighted as effective in detecting financial fraud, with specific methods showing varying degrees of success (accuracy, sensitivity, and specificity) across different types of fraud.
White (1995)	AI (ES)	Descriptive: describe the current state and opinions regarding the integration of ES and AI in accounting and auditing education	N.a	Quantitative analysis of survey data	There is a mismatch between what educators deem important ES and AI topics and the coverage of these topics in existing accounting information systems curriculum.
Widuri et al. (2016)	CAATTs (GAS)	Exploratory: explore the factors influencing the adoption of GAS at the firm level.	Technology, organisation, and environment framework	Qualitative analysis of interview data	Availability of information technology-skilled auditors in the local market, client needs and expectations and client size are important consideration for adopting GAS.
Will (1983)	CAATTs (GAS)	Design science: design and develop a specific language (Audit Command Language – ACL) intended for use in auditing.	MIS-auditor framework	Design science method	It details the specific functions and capabilities of ACL, including data audit commands, the ACL directory, and auxiliary functions, indicating how it supports auditors in their tasks.

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Table A2 (continued)

Authors (year)	Focal technology	Research typology and objectives	Main theoretical perspectives	Research methods	main findings
Wilson and Sharda (1994)	AI (MLNA)	Explanatory: examine the application of neural networks to predict bankruptcy and compare their performance with classical methods.	Machine learning theory	Quantitative analysis of archival data	Neural networks significantly outperform discriminant analysis in predicting both bankrupt and non-bankrupt firms under various conditions, reducing errors in this problem domain by as much as 93 % over chance.
Wongpinunwatana et al. (2000)	AI (ES)	Explanatory: examine the impact of task-technology fit on users' performance when using ES for auditing tasks	Task technology fit framework	Quantitative analysis of experimental data	It finds marginal improvements in problem-solving accuracy with case-based reasoning for unstructured tasks, no significant effect with rule-based reasoning for structured tasks, and a slight increase in users' certainty of their solutions
Yang et al. (2018)	AI (NLP)	Explanatory: examine the effectiveness of using NLP in mining corporate risk disclosure information to predict audit fee	Machine learning theory	Quantitative and qualitative analysis of archival data	Text mining using NLP is an effective alternative to understand and quantify risk of auditing clients.
Zhang et al. (2021)	AI (RPA)	Design science: propose and elaborate on an Attended Process Automation (APA) framework specifically tailored for the auditing context.	Technological process reframing theory	Design science method	The implementation of APA in auditing signifies a transformative shift towards enhancing audit efficiency and quality by enabling auditors to focus on judgment-intensive tasks while automating repetitive and rule-based activities.

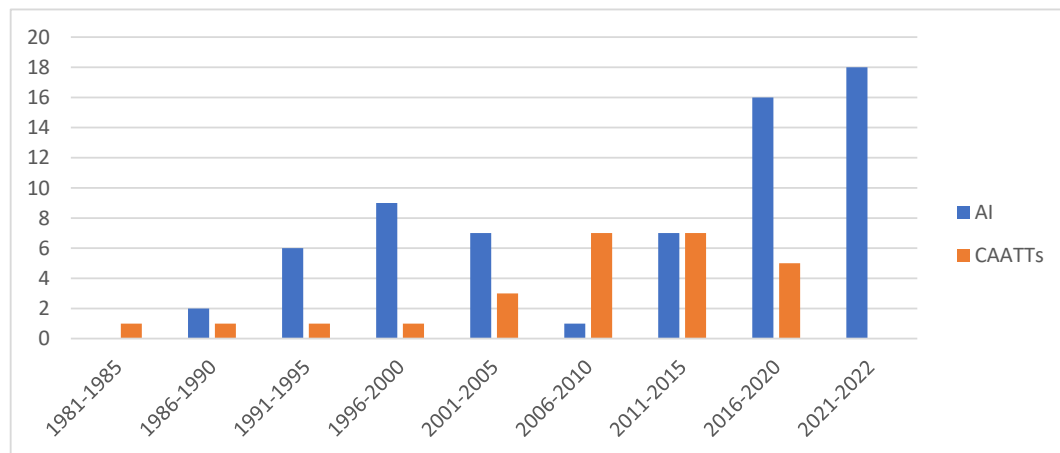


Fig. A1. Distribution of Articles Based on Year of Publication.

associations need to utilise AI for a transformative recalibration of their knowledge systems. By engaging in processes of knowledge creation, refinement, and retirement, these associations can maintain their professional legitimacy and relevance in an increasingly AI-driven landscape. Understanding the changing technological landscape and its strategic implications, actors within professional service can be better equipped to prepare for future trends, thus ensuring their relevance and sustainability in a rapidly evolving environment.

Although our review is anchored in an established theoretical framework and methodology, it is important to acknowledge some limitations. The first limitation pertains to the scope of our review, which is primarily focused on CAATTs and AI. While CAATTs serve as a representative group for classical IT applications in professional services, they do not encapsulate the entirety of classic IT tools. Additionally, AI technology is evolving and, as such, our review study is unable to address the effects of these emerging technologies not yet covered in existing literature at the time of writing this paper. For example, we anticipate that GenAI will emerge as a significant topic of interest in the future, considering its impact on knowledge and professional works (Alavi et al., 2024; Jarvenpaa & Klein, 2024; Schwartz & Te'eni, 2024). Therefore, we encourage other researchers to broaden the exploration to include various forms of classic IT and AI in future studies. The second potential limitation of our study is rooted in the theoretical framework we employed.

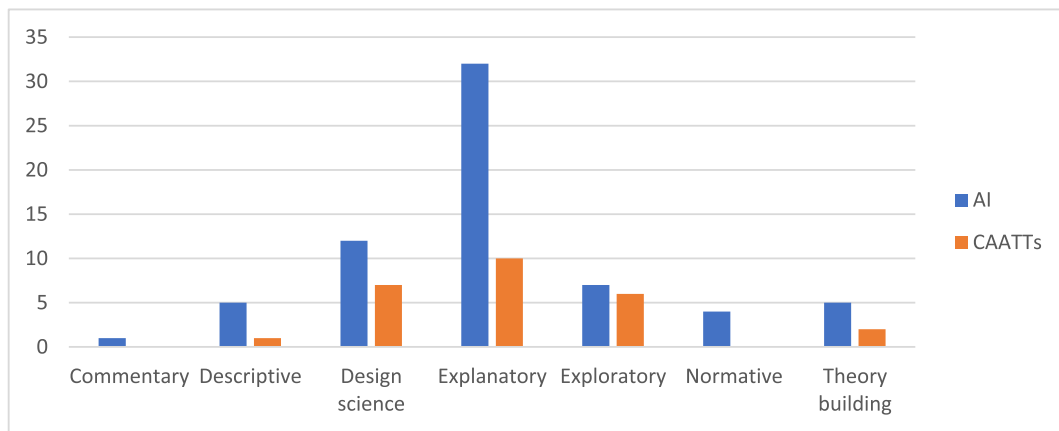


Fig. A2. Distribution of Articles Based on Research Types.

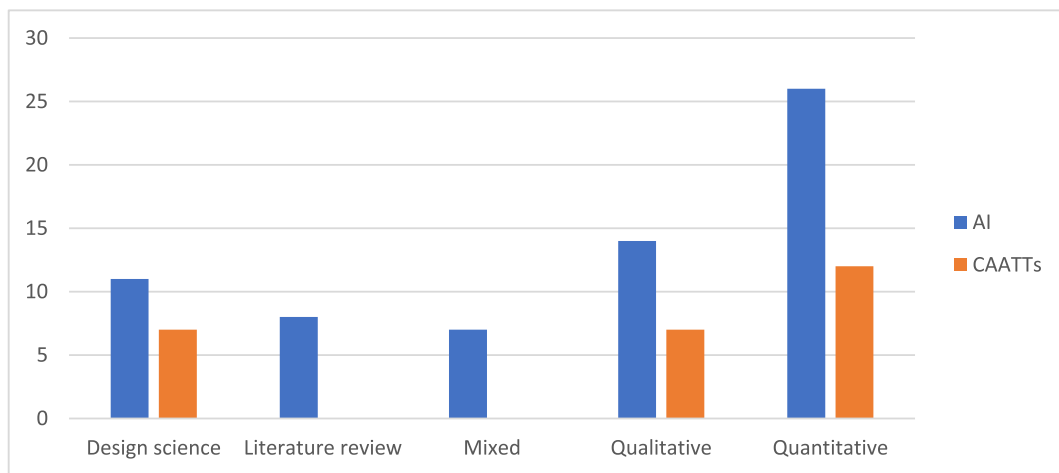


Fig. A3. Distribution of Articles Based Research Methods.

Our analysis is based on the TACT framework, which is designed to understand the goal orientation of actors in specific contexts. Thus, in identifying these goals, our approach relies heavily on discussions within the articles included in our literature review, rather than drawing on broader organisational theory. This reliance confines us to the theoretical lenses adopted in the reviewed papers, potentially overlooking other relevant goals. For instance, maintaining legitimacy might not only be a goal for professional associations but also become a goal for some large professional service firms. We have made some initial attempts to connect our findings to relevant organisational theory and literature. To further address this limitation, we encourage future studies to adopt different theoretical perspectives, which could provide a more comprehensive understanding of the dynamics at play.

CRedit authorship contribution statement

Jiaqi Yang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alireza Amrollahi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Conceptualization. **Mauricio Marrone:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

In this section, we present an overview of the distribution of articles reviewed, categorised by their focal technology, publication years, types of research, and the methodologies they employ. We then provide a comprehensive description and classification of each article. Table A1 showcases a significant emphasis on AI, particularly with 26 articles dedicated to Machine Learning Numerical Analytics. This is followed by Robotic Process Automation and Expert Systems, each represented in six articles. Interestingly, Computer Vision emerges as a relatively new and less explored field within AI-enabled auditing research. It is also noteworthy that nine studies explore a combination of AI technologies. These studies often provide an overview of AI applications in the auditing profession or combine different subcategories of AI to demonstrate their potential use and impact on auditing practices. Within CAATs, General Audit Software stands out with 15 articles, signifying its pivotal role in traditional audit techniques. Table A2 provides a summary of the review articles, including the author, year, focal technology, research typology and objectives, main theoretical perspectives, research methods, and key findings.

Fig. A1 illustrates the trends in publication over the years, marking a surge in AI-focused research starting in 2011, with a remarkable output of nearly 20 papers in just two years leading up to the review. This trend indicates that AI has drawn significant attention from scholars in recent years due to its profound implications in the auditing field. Conversely, interest in CAATs research appears to have diminished after reaching a peak in 2015, indicating a shift in scholarly focus from traditional IT tools to AI technologies in the auditing field. Figs. A2 and A3 expose the varied approaches employed in the study of AI and CAATs. Nearly half of AI research is explanatory and using quantitative methods, emphasising the aim of elucidating the mechanics and implications of AI technologies through empirical data and statistical analyses. Design science and qualitative exploratory research constitute the second and third most common approaches, respectively, in the investigation of AI applications within auditing. On the other hand, CAATs research exhibits a more balanced distribution of methodologies, including design science, qualitative exploratory, and quantitative explanatory studies.

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